

Genome concentration limits cell growth and modulates proteome composition in *Escherichia coli*

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Abstract

Defining the cellular factors that drive growth rate and proteome composition is essential for understanding and manipulating cellular systems. In bacteria, ribosome concentration is known to be a constraining factor of cell growth rate, while gene concentration is usually assumed not to be limiting. Here, using single-molecule tracking, quantitative single-cell microscopy, and modeling, we show that genome dilution in *Escherichia coli* cells arrested for DNA replication results in a decrease in the concentration of active RNA polymerases and ribosomes. The resulting sub-linear scaling of total active RNA polymerases and ribosomes with cell size leads to sub-exponential growth, even within physiological cell sizes. Cell growth rate scales proportionally with the total number of active ribosomes in a DNA concentration-dependent manner. Tandem-mass-tag mass spectrometry experiments further reveal that a decrease in DNA-to-cell-volume ratio proportionally remodels the composition of the proteome with cell size independently of the environment. Altogether, our findings indicate that genome concentration is an important driver of exponential cell growth and a global modulator of proteome composition in *E. coli*. Comparison with studies on eukaryotic cells suggests DNA concentration-dependent scaling principles of gene expression across domains of life.

eLife assessment

This study represents a **fundamental** contribution to our understanding of how gene expression levels are controlled in bacteria. Through a series of **compelling** and careful experiments, relying on a mutant that blocks DNA replication but permits growth, and using various methods, the authors reveal how genome concentration rapidly becomes limiting for growth when replication is inhibited. This work contributes to our understanding of the contributions and limiting roles of DNA, mRNA, and ribosomes for growth in bacteria, and will be of considerable interest within both systems biology and microbial physiology.

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Introduction

Cells regulate the intracellular concentration of various proteins and macromolecules to modulate the rate of essential cellular processes, including growth. In bacteria, cell mass and volume typically double between division cycles. Proportionality between biosynthetic capacity and biomass accumulation results in exponential or near-exponential cell growth during the cell cycle (Campos et al., 2014 [↗](#); Schaechter et al., 1958 [↗](#); Siegal-Gaskins and Crosson, 2008 [↗](#); Taheri-Araghi et al., 2015 [↗](#); Wang et al., 2010 [↗](#)). What drives exponential growth has been a longstanding question in the microbiology field (Belliveau et al., 2021 [↗](#); Churchward et al., 1982 [↗](#); Ecker and Schaechter, 1963 [↗](#); Zhurinsky et al., 2010 [↗](#)). Quantitative studies on model bacteria such as *Escherichia coli* place the concentration of ribosomes and their kinetics as the principal rate-limiting factors (Belliveau et al., 2021 [↗](#); Bosdriesz et al., 2015 [↗](#); Koch, 1988 [↗](#); Scott et al., 2014 [↗](#), 2010 [↗](#)). Most other cellular components essential for growth are estimated to be at least an order of magnitude above the level required for proper enzymatic reactions (Belliveau et al., 2021 [↗](#)), indicating that they are well in excess in terms of metabolic concentrations. Thus, translation is generally seen as the rate-governing process for cellular growth. While the translocation rate of ribosomes poses an inherent limit on the growth rate of the cell, protein concentrations are predominantly set transcriptionally at the promoter level, with tight coordination between transcription and translation (Balakrishnan et al., 2022 [↗](#)).

Whereas the importance of ribosome concentration in growth rate determination has been extensively studied, a potential role for genome concentration has received less attention. An early population study on an *E. coli* thymine auxotroph proposed that global transcription is not limited by the concentration of the genome but is instead constrained by the availability of RNA polymerases (RNAPs) (Churchward et al., 1982 [↗](#)). However, the potential impact of DNA concentration on determining the growth rate of *E. coli* or other bacteria has, to our knowledge, not been formally tested. Interestingly, *E. coli* and *Bacillus subtilis* have been shown to display small but reproducible deviations from exponential growth during the division cycle (Kar et al., 2021 [↗](#); Nordholt et al., 2020 [↗](#)), with the growth rate increasing after the initiation of DNA replication under some conditions. Furthermore, at the population level, these organisms initiate DNA replication at a fixed cell volume (mass) per chromosomal origin of replication (*oriC*) across a wide range of nutrient and genetic conditions (Donachie, 1968 [↗](#); Govers et al., 2024 [↗](#); Si et al., 2017 [↗](#); Zheng et al., 2016 [↗](#)), suggesting that DNA concentration is an important physiological parameter for these bacteria. In eukaryotes where genome concentration is also tightly controlled (Ginzberg et al., 2015 [↗](#); Turner et al., 2012 [↗](#)), a change in DNA-to-cell-volume ratio has recently

been demonstrated to remodel the proteome and promote cellular senescence (Crozier et al., 2023 [↗](#); Foy et al., 2023 [↗](#); Lanz et al., 2023 [↗](#), 2022 [↗](#); Manohar et al., 2023 [↗](#); Neurohr et al., 2019 [↗](#); Wilson et al., 2023 [↗](#)).

In this study, we combined single-cell and single-molecule microscopy experiments with tandem-mass-tag mass spectrometry and modeling to investigate the potential physiological role of genome concentration in cell growth and proteome composition in *E. coli*.

Results

Growth rate correlates with the genome copy number in *E. coli*

To examine the potential effect of DNA content on the growth rate of *E. coli*, we used two CRISPR interference (CRISPRi) strains with arabinose-inducible control of expression of dCas9 (Li et al., 2016 [↗](#); Si et al., 2017 [↗](#)). One strain expressed a single guide RNA (sgRNA) against *oriC* where sequestration by dCas9 binding prevents the initiation of DNA replication, producing cells with a single copy of the chromosome (**Figure 1A** [↗](#)) (Si et al., 2017 [↗](#)). These cells, referred to as “1N cells” below, grew into filaments as a block in DNA replication prevents cell division, but not cell growth, from occurring (Carl, 1970 [↗](#); Si et al., 2017 [↗](#); Withers and Bernander, 1998 [↗](#)). The second CRISPRi strain, which served as a comparison, expressed a sgRNA against the cell division protein FtsZ. FtsZ depletion blocks cell division while allowing DNA replication to proceed (**Figure 1A** [↗](#)) (Addinall et al., 1996 [↗](#); Li et al., 2016 [↗](#)). Ongoing growth resulted in filamenting cells with multiple replicating chromosomes, hereafter referred to as “multi-N cells”. For both strains, we used time-lapse microscopy to monitor growth at the single-cell level at 37°C in M9 minimal medium supplemented with glycerol, casamino acids, and thiamine (M9glyCAAT). Cell area (*A*) was automatically detected from phase-contrast images using a deep convolutional network (Wiktor et al., 2021 [↗](#)), and the absolute growth rate ($\frac{dA}{dt}$) was determined by calculating the difference in cell area between frames. The relative growth rate ($\frac{1}{A} \frac{dA}{dt}$), which is constant for exponential growth, was calculated by dividing by the cell area. We chose to use cell area rather than cell volume for growth rate measurements because cell area is what we measured in our two-dimensional cell images, and the average cell width remained largely constant for 1N and multi-N cells following CRISPRi induction (**Figure 1 – figure supplement 1A** [↗](#)). Furthermore, we found that absolute and relative growth rates based on extracted cell volumes produced similar results (**Figure 1 – figure supplement 1B-K** [↗](#)) to those based on cell areas (**Figure 1** [↗](#) and **Figure 1 – figure supplement 2** [↗](#)). Therefore, we will report growth rate measurements based on cell areas hereafter.

For the CRISPRi *oriC* strain, we selected 1N cells that contained a single DNA object (nucleoid) labeled by a mCherry fusion to the nucleoid-binding protein HupA (referred to as HU below). To confirm this 1N chromosome designation, we used a CRISPRi *oriC* strain that expresses HU-CFP and carries an *oriC*-proximal *parS* site labeled with ParB-mCherry (**Figure 1 – figure supplement 3A** [↗](#)), used here to determine the number of nucleoids and chromosomal origins per cell. We found that $96 \pm 1\%$ (mean $\pm$ standard deviation, SD, *n* = 3378 cells, three biological replicates) of cells with a single HU-labeled nucleoid contained no more than one ParB-mCherry focus, indicative of a single *oriC*.

Using this methodology, we observed a significant difference in growth rate between 1N and multi-N cells as shown in representative single-cell growth trajectories (**Figure 1B** [↗](#)) and in aggregated absolute growth rate measurements (**Figure 1C** [↗](#)). In multi-N cells, the absolute growth rate rapidly increased with cell area. For these cells, the scaling between absolute growth rate and cell size might even be slightly greater than expected for exponential growth, as the relative growth rate was not perfectly constant and mildly increased with cell area (**Figure 1E** [↗](#)). This slight increase would be consistent with the super-exponential growth pattern previously

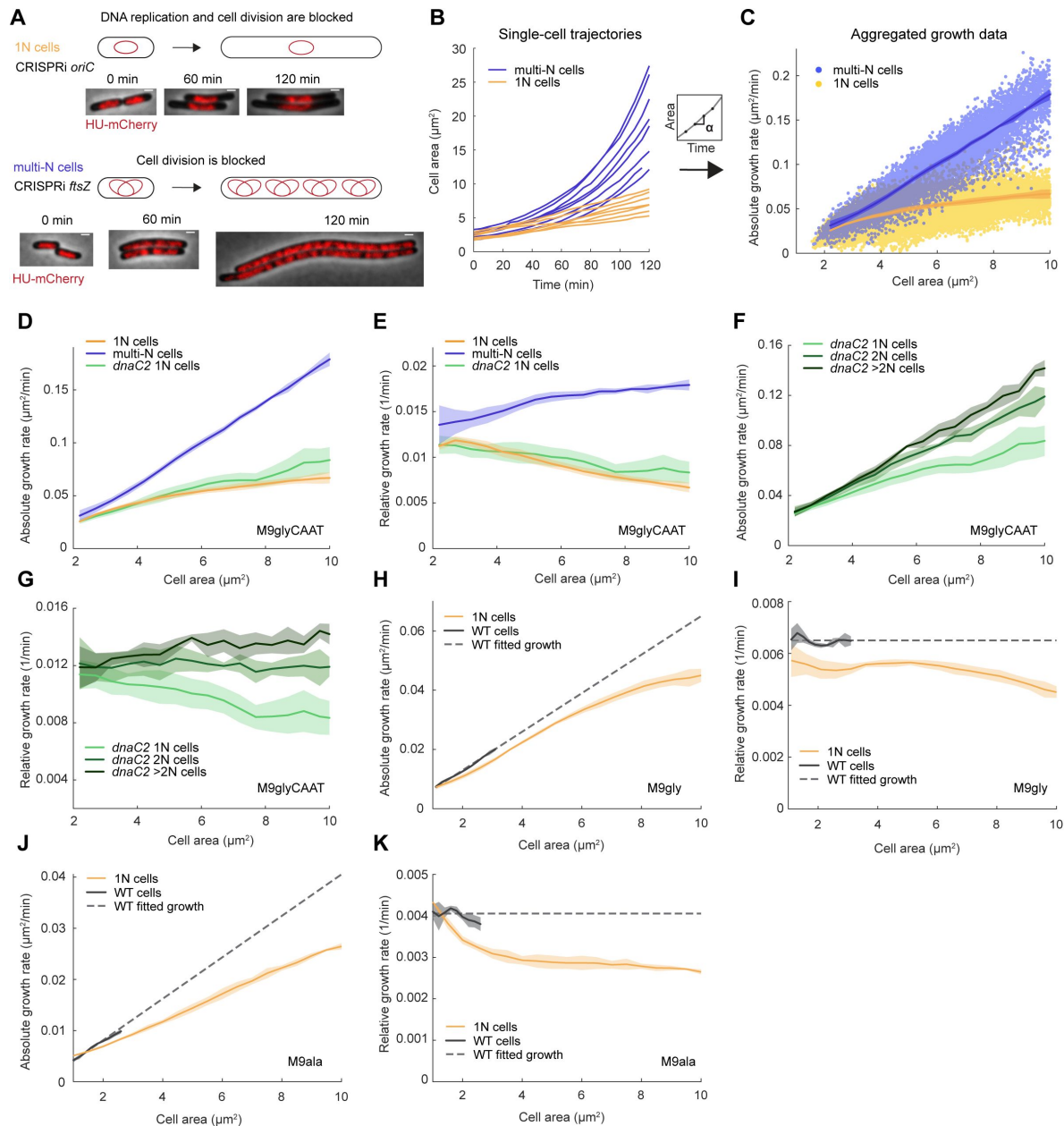


Figure 1.

Growth rate and genome copy number in *E. coli*.

(A) Illustration of 1N (CRISPRi *oriC*, CJW7457) and multi-N (CRISPRi *ftsZ*, CJW7576) cells with different numbers of chromosomes along with representative microscopy images at different time points following CRISPRi induction. Scale bars: 1 μm . (B) Plot showing representative single-cell trajectories of cell area as a function of time for the CRISPRi strains following a block in DNA replication and/or cell division. (C) Plot showing the absolute growth rate as a function of cell area for 1N (32735 datapoints from 1568 cells) and multi-N cells (14006 datapoints from 916 cells) in M9glyCAAT. Lines and shaded areas denote mean $\pm$ SD from three experiments. This also applies to the panels below. (D) Absolute and (E) relative growth rate in 1N (32735 datapoints from 1568 cells, CJW7457), multi-N (14006 datapoints from 916 cells, CJW7576), and *dnaC2* 1N (13933 datapoints from 1043 cells, CJW7374) cells as a function of cell area in M9glyCAAT. (F) Absolute growth rate and (G) relative growth rate in 1N (13933 datapoints from 1043 cells), 2N (6265 datapoints from 295 cells) and >2N (2116 datapoints from 95 cells) *dnaC2* (CJW7374) cells as a function of cell area in M9glyCAAT. (H) Absolute and (I) relative growth rate in 1N (50352 from 973 cells, CJW7457) and WT (80269 datapoints from 12544 cells, CJW7339) cells in M9gly. (J) Absolute and (K) relative growth rate in 1N (71736 datapoints from 909 cells, CJW7457) and WT (63367 datapoints from 6880 cells, CJW7339) cells in M9ala. Lines and shaded areas denote mean $\pm$ SD from three biological replicates.

described for wild-type (WT) cells (Cylke and Banerjee, 2023 [↗](#); Kar et al., 2021 [↗](#)). Multi-N cells grew identically to WT within the same cell size range (**Figure 1 – figure supplement 2** [↗](#)). In contrast, 1N cells grew significantly slower than multi-N-cells and their absolute growth rate only moderately increased with cell area, approaching an apparent plateau at large cell sizes (**Figure 1C-D** [↗](#)). This sub-exponential growth resulted in a relative growth rate that decreased with cell area (**Figure 1E** [↗](#)). The striking divergence in growth between 1N and multi-N cells of the same size suggested that growth is dependent on DNA concentration. The difference in growth rate between 1N and multi-N cells was already apparent in the physiological range of cell sizes when compared to WT cells (**Figure 1 – figure supplement 2** [↗](#)), suggesting that growth rate reduction occurs soon after DNA replication fails to initiate.

We ruled out that the slower growth of 1N cells was due to an induction of the SOS response or to an increased level in the nucleotide alarmone (p)ppGpp, as inactivation of either stress pathway (through deletion of *recA* or *spoT/relA*, respectively) in 1N cells made little to no difference to their growth rate (**Figure 1 – figure supplement 4** [↗](#)). Furthermore, as an independent validation, we used an orthogonal system to block DNA replication using the temperature-sensitive mutant *dnaC2*, which encodes a deficient DNA helicase loader at the restrictive temperature of 37°C (Carl, 1970 [↗](#); Withers and Bernander, 1998 [↗](#)). We observed that the relationship between absolute growth rate and cell area in *dnaC2* cells with a single nucleoid was similar to that of 1N cells produced by the CRISPRi *oriC* system (**Figure 1D and E** [↗](#)).

We noticed that, even at the restrictive temperature, the *dnaC2* strain produced a sizeable fraction of cells with more than one HU-mCherry-labelled nucleoid (**Figure 1 – figure supplement 3B** [↗](#) and C), indicating that the temperature-sensitive effect on DNA replication is not fully penetrant. We took advantage of this phenotypic “leakiness” to measure the growth rate of cells with different numbers of nucleoids (and thus chromosomes) within the *dnaC2* population. We observed a notable difference in growth rate between cells of 1, 2, and >2 nucleoids in the population, with each additional nucleoid contributing to higher cellular growth at a given cell size (**Figure 1F-G** [↗](#)). This finding is consistent with DNA-limited growth in which cellular growth rate increases with genome concentration.

In the relatively nutrient-rich M9glyCAAT condition, WT cells at birth are expected to have higher DNA content than 1N cells on average due to overlapping DNA replication (Fossum et al., 2007 [↗](#)). This difference in DNA concentration explains the lower absolute growth rate of 1N cells relative to WT even at the smallest cell sizes (**Figure 1 – figure supplement 2** [↗](#)). To examine whether cells are also subject to DNA-limited growth when multi-fork DNA replication is rare or nonexistent, we measured growth of 1N cells in two different nutrient-poor media, M9 glycerol (M9gly) and M9 L-alanine (M9ala). In both conditions, the growth rate scaled slower with cell area in 1N cells relative to WT cells (**Figure 1H-K** [↗](#)). Here again, this difference in growth was apparent in the cell size range of WT cells, indicating that the difference in growth rate is not simply the result of a non-physiological regime. Altogether, our data suggest that in WT cells, the genome concentration is (or is close to be) limiting for growth and that growth rate scales with ploidy.

The concentration of ribosomal proteins remains relatively constant in DNA-diluted cells

Ribosome content is often proposed to explain the approximately exponential growth of biomass in bacteria, with growth rate being directly proportional to ribosome concentration (Bremer and Dennis, 1996 [↗](#); Ecker and Schaechter, 1963 [↗](#); Scott et al., 2014 [↗](#), 2010 [↗](#)). Therefore, we first quantified the fluorescence concentration of a monomeric superfolder green fluorescent protein (msfGFP) fusion to the ribosomal protein RpsB (expressed from the native chromosomal locus) in 1N and multi-N cells in M9glyCAAT as a function of cell area. We found it to be almost identical between the two CRISPRi strains and relatively constant across cell areas, regardless of DNA content (**Figure 2A** [↗](#)).

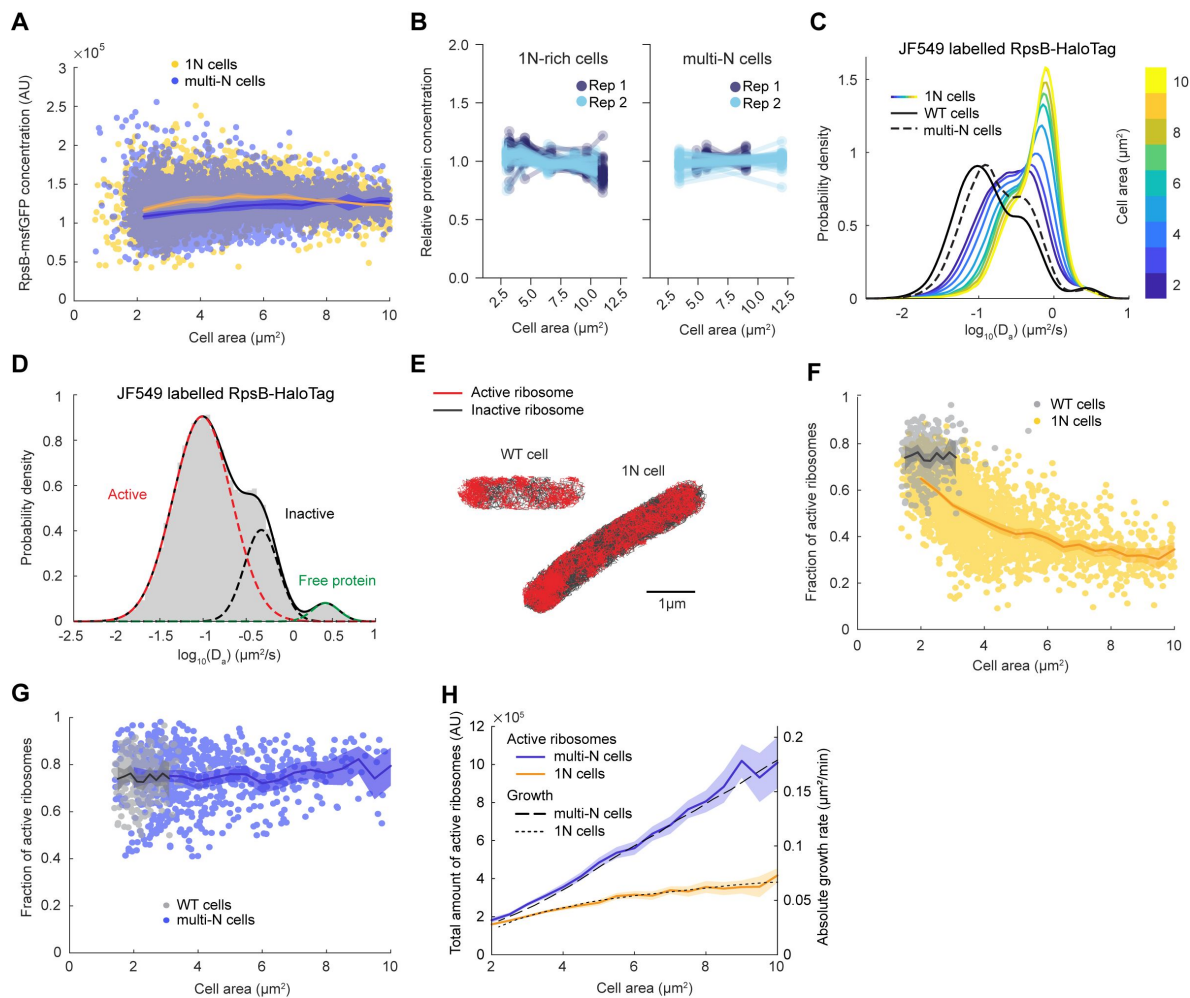


Figure 2.

Lower ribosome activity explains the reduced growth rate of 1N cells.

(A) RpsB-msfGFP fluorescence concentration in 1N (6542 cells, CJW7478) and multi-N (10537 cells, CJW7564) cells as a function of cell area. Lines and shaded areas denote mean $\pm$ SD from three experiments. (B) Relative protein concentration of different ribosomal proteins in 1N (SJ_XTL676) and multi-N (SJ_XTL229) cells by TMT-MS. 1N-rich cells were collected 0, 120, 180, 240 and 300 min after addition of 0.2% arabinose, while multi-N cells were collected after 0, 60, and 120 min of induction. Blue and cyan represent two independent experiments. Only proteins with at least four peptide measurements are plotted. (C) Apparent diffusion coefficients (D_a) of JF549-labelled RpsB-HaloTag in WT (32,410 tracks from 771 cells, CJW7528), 1N (848,367 tracks from 2478 cells, CJW7529) and multi-N cells (107,095 tracks from 1139 cells, CJW7530). Only tracks of length ≥ 9 displacements are included. 1N cells are color-binned according to their cell area while multi-N cells contain aggregated data for ~ 2 - $10 \mu\text{m}^2$ cell areas. (D) D_a in WT cells fitted by a three-state Gaussian mixture model (GMM): $77 \pm 1\%$, $20 \pm 1\%$, and $3.2 \pm 0.5\%$ ($\pm$ SEM) of the ribosome population, from the slowest moving to the fastest moving (32,410 tracks from 771 cells). (E) Example WT and 1N cells where (red, slow-moving) active and (gray, fast-moving) inactive ribosomes are classified according to the GMM. (F) Active (slow-moving) ribosome fraction in individual WT (237 cells) and 1N (2453 cells) cells as a function of cell area. Only cells with ≥ 50 tracks are included. Lines and shaded areas denote mean and 95% confidence interval (CI) of the mean from bootstrapping. (G) Same as (F) but for WT (237 cells) and multi-N (683 cells) cells. (H) Absolute growth rate of 1N and multi-N cells (Figure 1C) as a function of cell area was overlaid with the total active ribosome amount (calculated from Fig. 2A, F and G). Lines and shaded areas denote mean and 95% CI of the mean from bootstrapping.

To exclude the possibility that the msfGFP tag altered the synthesis of RpsB or that this protein behaved differently from other ribosomal proteins, we adapted a tandem-mass-tag (TMT) mass spectrometry (MS) method recently developed to examine cell size-dependent proteome scaling in yeast and human cells (Lanz et al., 2022 [DOI](#)). Note that for the CRISPRi *oriC* strain, a minority (~10–15%) of cells have more than one nucleoid. These cells were excluded from the analysis of our single-cell microscopy experiments. However, this could not be done for the TMT-MS experiments, which provide population-level measurements. Therefore, for this TMT-MS section, we will refer to the CRISPRi *oriC* cell population as “1N-rich” cells, instead of just “1N” cells. Using the TMT-MS approach, we found that the relative concentration of all (54) high-abundance ribosomal proteins (including untagged RpsB) remained approximately constant across all sizes of 1N-rich cells and was relatively similar between 1N-rich and multi-N cells (**Figure 2B** [DOI](#)). Only the ribosomal protein L31B, a stationary phase paralog of the more prevalent exponential phase ribosomal protein L31A (Lilleorg et al., 2019 [DOI](#)), displayed significant dilution in 1N cells (Dataset 1). Ribosomal protein concentration also appeared to remain constant in 1N cells in nutrient-poor media based on RpsB-msfGFP fluorescence intensity measurements in single cells (**Figure 2 – figure supplement 1** [DOI](#)). Thus, the concentration of ribosomal proteins does not explain the difference in growth rate between cells with different ploidy.

The fraction of active ribosomes is reduced in DNA-diluted cells

To more specifically probe the translational activity of ribosomes in 1N cells, we performed single-molecule tracking in live cells growing in M9glyCAAT. Ribosomes are expected to exhibit at least two different dynamic states: slow mobility when active (i.e., engaged in translation on the mRNA, often in polyribosome form), and faster mobility when inactive ribosomes (or ribosomal subunits) are diffusing in the cytoplasm (Mohapatra and Weisshaar, 2018 [DOI](#); Sanamrad et al., 2014 [DOI](#)). To track ribosomes, we introduced a HaloTag fusion to RpsB at the endogenous locus and labeled the HaloTag using the membrane-permeable Janelia Fluor 549 (JF549) fluorescent dye (Grimm et al., 2015 [DOI](#)). We quantified the apparent diffusion coefficient (D_a) of single-molecule tracks in WT cells as well as in 1N and multi-N cells at multiple time points following CRISPRi induction (**Figure 2C** [DOI](#)). We found that the distribution of D_a in multi-N cells of all sizes (~2–10 μm^2) was similar to that in WT cells despite the considerable differences in cell sizes. In contrast, 1N cells displayed distributions clearly distinct from WT and multi-N cells, gradually shifting toward faster mobilities (higher D_a) with increasing cell size. This shift suggests that ribosome activity is altered in 1N cells.

Gaussian fitting of the D_a logarithmic data in WT cells revealed two predominant dynamic states of ribosomes: a slow-diffusing and a fast-diffusing state, representing $77 \pm 1\%$ (mean $\pm$ standard error of the mean (SEM)) and $20 \pm 1\%$ of the ribosome population, respectively (**Figure 2D** [DOI](#)). In addition, we observed a small fraction ($3.2 \pm 0.5\%$) of faster-moving molecules with D_a expected for freely-diffusing proteins (Banaz et al., 2019 [DOI](#); Elowitz et al., 1999 [DOI](#)), likely indicative of a small pool of free RpsB-HaloTag proteins (i.e., not assembled into ribosomes). To confirm that the slow-diffusing fraction corresponded to translationally active ribosomes, we showed that this fraction nearly vanished (down to $1.10 \pm 0.02\%$) when cells were depleted of mRNAs following 30 min treatment with the transcription inhibitor rifampicin (**Figure 2 – figure supplement 2** [DOI](#)). The estimated fraction (~77%) of active ribosomes in untreated cells was in good agreement with previous single-molecule and biochemical studies under these growth conditions (Blundell and Wild, 1971 [DOI](#); Mohapatra and Weisshaar, 2018 [DOI](#); Phillips et al., 1969 [DOI](#); Sanamrad et al., 2014 [DOI](#)).

Upon fitting the D_a values of ribosomes in 1N cells of different sizes (**Figure 2E** [DOI](#)), we observed a significant reduction in the slow-diffusing ribosome population (**Figure 2 – figure supplement 3** [DOI](#)). Quantification of the active (slow-diffusing) ribosome fraction per cell revealed that 1N cells have overall lower ribosome activity than WT cells, and that ribosome activity decreases monotonically with increasing cell area (**Figure 2F** [DOI](#)). In contrast, ribosome activity in multi-N cells remained the same as in WT across different cell sizes (**Figure 2G** [DOI](#)). The reduction in

fraction of active ribosomes was less pronounced in 1N cells growing in M9gly and M9ala (**Figure 2 – figure supplement 4**), consistent with the smaller deviation in growth rate between 1N and WT cells (**Figure 1H-K**).

To estimate the total number of active ribosomes per cell in M9glyCAAT, we multiplied the total amount of ribosomes by the fraction of active ribosomes and plotted the result as a function of cell area (**Figure 2H**). We found that the difference in the total number of active ribosomes between 1N and multi-N cells matches the observed difference in growth rate in M9glyCAAT (**Figure 2H**), indicating cell growth rate is directly proportional to the increase in total active ribosomes. We also estimated the total number of active ribosomes in WT and 1N cells in M9gly and M9ala, where, overall, we observed similar trends (**Figure 2 – figure supplement 4G-H**). Altogether, the results are consistent with the hypothesis that DNA limitation decreases ribosome activity, which in turn reduces the growth rate.

DNA dilution reduces the activity of RNA polymerases

We reasoned that the observed changes in ribosome activity in 1N cells may reflect the available mRNA pool in cells. If true, we would expect the total activity of RNAPs to be reduced in 1N cells. In cells, the total activity of RNAPs is determined by the concentration of RNAPs multiplied by the fraction of active RNAPs. Therefore, we first determined whether RNAP concentration was lower in 1N cells relative to multi-N cells in M9glyCAAT by quantifying the fluorescence intensity of a functional fusion of YFP to the RNAP β' subunit (encoded by *rpoC*) expressed from its native chromosomal locus. As expected, RNAP concentration remained constant in multi-N cells (**Figure 3A**). In 1N cells, the RNAP concentration increased with cell size (**Figure 3A**), the opposite of what would be expected to explain the growth rate defect. We confirmed this increasing trend in concentration for other protein subunits of the core RNAP and the primary sigma factor σ^{70} (encoded by *rpoD*) using TMT-MS (**Figure 3B**), clearly demonstrating that the abundance of RNAPs was not the limiting factor. RNAP concentration also increased, albeit to a lesser degree, in 1N cells in nutrient-poor media based on fluorescence intensity measurements (**Figure 3 – figure supplement 1**).

To quantify RNAP activity in 1N and multi-N cells, we performed single-molecule tracking in live cells in M9glyCAAT using a functional fusion of HaloTag to the β' protein subunit RpoC labeled with the JF549 dye. As expected, the D_a values of RpoC-HaloTag in multi-N cells were distributed similarly to those in WT cells (**Figure 3C**). In contrast, the distribution in 1N cells changed gradually toward higher D_a values (faster mobility) with increasing cell size (**Figure 3C**).

As with ribosomes, RNAPs primarily exhibited two major states of diffusivity (**Figure 3D**): a slower-diffusing fraction ($49 \pm 4\%$; mean $\pm$ SEM) and a faster-diffusing fraction ($49 \pm 4\%$), likely representing transcriptionally active RNAPs and inactive, diffusing RNAPs, respectively. A small fraction of RpoC-HaloTag ($2 \pm 0.1\%$) diffused very fast, with D_a values expected for free proteins, suggesting that it reflects the few β' proteins not assembled into the RNAP core complex. Using rifampicin treatment, we confirmed that the slowest state corresponded to RNAPs actively engaged in transcription (**Figure 3 – figure supplement 2**). In these rifampicin-treated cells, the slow-diffusing fraction was reduced to $13 \pm 4\%$. Rifampicin does not prevent promoter binding or open complex formation and instead blocks transcription elongation following the synthesis of 3-nucleotide-long RNAs (Campbell et al., 2001). Thus, the observation that slow-moving RNAPs did not completely disappear after rifampicin treatment is consistent with the mechanism of action of the drug, leaving a fraction of RNAPs bound at promoter sites.

Unlike in WT and multi-N cells, the fraction of active RNAPs in 1N cells decreased monotonically with increasing cell area in all tested media (**Figure 3F-G** and **Figure 3 – figure supplements 3** and **4**). However, because the RNAP concentration simultaneously increased in 1N cells, it was possible that the total amount of active RNAP, which is the relevant metric of transcription activity, remained equal to that of multi-N cells. By calculating the total amount of active RNAPs,

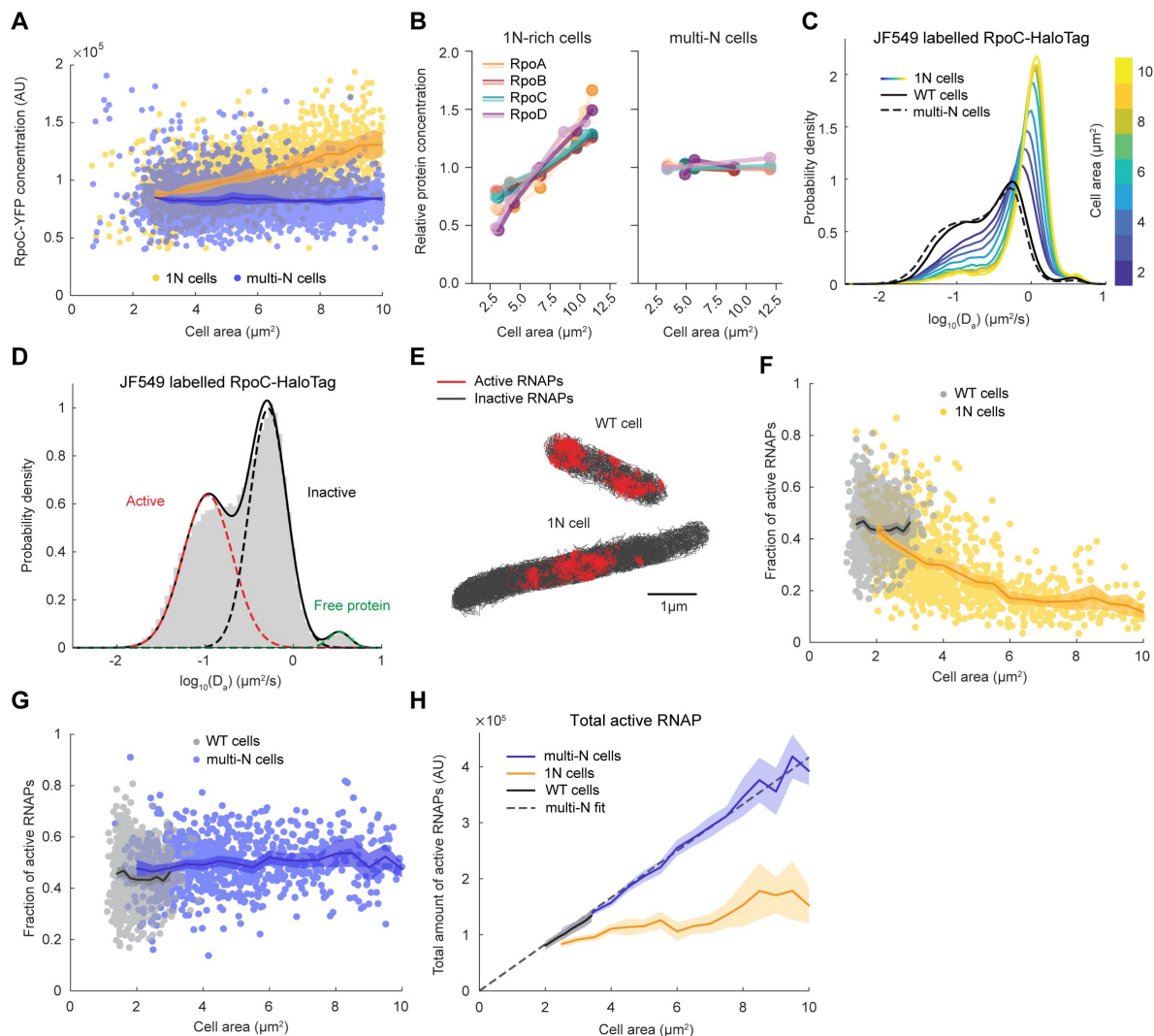


Figure 3.

RNAP activity is reduced in 1N cells.

(A) RpoC-YFP fluorescence concentration in 1N (3580 cells, CJW7477) and multi-N (5554 cells, CJW7563) cells as a function of cell area. Lines and shaded areas denote mean $\pm$ SD from 3 experiments. (B) Relative protein concentration of core RNAP subunits and σ^{70} in 1N-rich (SJ_XTL676) and multi-N (SJ_XTL229) cells by TMT-MS. 1N-rich cells were collected 0, 120, 180, 240 and 300 min after addition of 0.2% L-arabinose, while multi-N cells were collected after 0, 60, and 120 min of induction. (C) Apparent diffusion coefficients of JF549-labelled RpoC-HaloTag in WT (91,280 tracks from 1000 cells, CJW7519), 1N (175,884 tracks from 1219 cells, CJW7520) and multi-N cells (186,951 tracks from 1040 cells, CJW7527). Only tracks of length ≥ 9 displacements are included. 1N cells are binned according to cell area while multi-N cells contain aggregated data for ~ 2 - $15 \mu\text{m}^2$ cell areas. (D) D_a in WT cells fitted by a 3-state GMM: $49 \pm 4\%$, $49 \pm 4\%$, and $2 \pm 0.1\%$ ($\pm$ SEM) of the RNAP population, from the slowest moving to the fastest moving (91,280 tracks from 1,000 cells). (E) Example WT and 1N cells where (red, slow-moving) active and (gray, fast-moving) inactive RNAPs are classified according to the GMM. (F) Active RNAP fraction in individual WT (854 cells) and 1N (1024 cells) cells as a function of cell area. Only cells with at least 50 tracks are included. Lines and shaded areas denote mean $\pm$ 95% CI of the mean from bootstrapping (3 experiments). (G) Same as (F) but for WT (854 cells) and multi-N (924 cells) cells. (H) Total amount of active RNAP in WT, 1N and multi-N cells as a function of cell area (calculated from Fig. 3A, F and G). Also, shown is a linear fit to multi-N data ($f(x) = 4.16 \cdot 10^{-4} \cdot x$, R^2 0.98). Lines and shaded areas denote mean and 95% CI of the mean from bootstrapping. All microscopy data are from three biological replicates.

we showed that the decrease in active fraction in 1N cells was not the mere result of the increase in RNAP concentration. Indeed, in M9glyCAAT, the total amount of active RNAPs hardly increased with cell size in 1N cells whereas it increased proportionally with cell size in both multi-N and WT cells (**Figure 3H** [↗](#)). This was also true in M9gly and M9ala (**Figure 3 – figure supplement 4G-H** [↗](#)). The reduction in the total amount of active RNAPs was also readily apparent in 1N cells within the range of WT cell sizes (**Figure 3H** [↗](#) and **Figure 3 – figure supplement 4G-H** [↗](#)).

A recent study has shown that the intracellular concentration of Rsd, the anti-sigma factor of σ^{70} , increases in WT cells under slower growth conditions, causing a reduction in global mRNA synthesis (Balakrishnan et al., 2022 [↗](#)). Therefore, we verified that the concentration of Rsd remains approximately constant in both 1N-rich and multi-N cells based on our TMT-MS data (**Figure 3 – figure supplement 5** [↗](#)), eliminating Rsd as possible source of reduced RNAP activity in 1N cells. Instead, our data supports the notion that substrate (DNA) limitation leads to a reduced transcription rate, which would reduce the pool of mRNAs available for ribosomes.

DNA dilution can result in sub-exponential growth through mRNA limitation

In a previous theoretical study, Lin and Amir (2018) [↗](#) considered distinct scenarios for gene expression. Their model predicted that if DNA and mRNAs are in excess, cells display exponential growth, whereas if DNA and mRNAs are limiting, cells will adopt linear growth. Our experiments showed that 1N cells converge toward linear growth (toward slope = 0 in **Figure 1C, H** [↗](#), and **J**), though the complete transition to linear growth required a large decrease in DNA concentration. To quantitatively examine this transition from exponential to linear growth through DNA dilution, we developed two deterministic ordinary differential equation (ODE) models of the flow of genetic information that include parameters for the fractions of active RNAPs and ribosomes. In these models, the dynamics of mRNA (X) and protein (Y) numbers in the cell are described by

$$\frac{dX}{dt} = r_1 \alpha_{RNAP}(X, Y) Y - \delta X \quad (\text{Eq.1})$$

$$\frac{dY}{dt} = r_2 \alpha_{ribo}(X, Y) Y \quad (\text{Eq.2})$$

Where r_1 is the bulk transcription rate normalized by the total protein number, r_2 is the bulk translation rate normalized by the total protein number, $\alpha_{RNAP}(X, Y)$ is the fraction of active RNAPs, δ , is the mRNA degradation rate, $\alpha_{ribo}(X, Y)$ is the fraction of active ribosomes, and $\frac{dY}{dt}$ corresponds to the absolute growth rate. For simplicity, we assumed that the protein degradation is negligible and that the cell volume and the number of rRNAs grow proportional to protein Y (Balakrishnan et al., 2022 [↗](#); Lin and Amir, 2018 [↗](#)). For detailed description and estimation of the model parameters, see **Table 1** [↗](#) and Appendix 1.

Based on the function form of $\alpha_{RNA}(X, Y)$, we consider two ODE model variants. Model A examined the effect of DNA limitation with minimal mathematical complexity by assuming that the proteome does not change (see Materials and Methods). Model B considered RNAP kinetics (with three different RNAP states: free, promoter-bound, and transcribing) and took into consideration the experimentally observed increase in RNAP concentration in 1N cells (**Figure 3A-B** [↗](#), see Materials and Methods and Appendix 2). For both models, α_{RNAP} depended on DNA concentration. In 1N cells, the DNA amount was fixed to 1 genome while it scaled with cell volume in multi-N cells. We then used both models to perform simulations and compared the results to our measurements, starting with parameter values extracted or estimated from the *E. coli* literature (**Table 2** [↗](#)). The parameters were then optimized to fit six experimental datasets simultaneously: cell growth rate, the fraction of active RNAPs, and the fraction of active ribosomes in both 1N and multi-N cells in M9glyCAAT (see Materials and Methods and Appendix 3).

Table 1

Model parameter description

Parameter	Biological meaning	Unit	Description
r_1	$\frac{\text{bulk transcription rate}}{\text{total protein number}}$	$10^{-3}/\text{min}$	This parameter was estimated as $r_1 = \theta_{RNAP} \beta_{mRNA} / T_{RNAP}$, where θ_{RNAP} is the number of RNAPs normalized by the number of proteins; β_{mRNA} is the fraction of RNAPs engaged in transcription, and T_{RNAP} is the mean mRNA synthesis time.
r_2	$\frac{\text{bulk translation rate}}{\text{total protein number}}$	$10^{-3}/\text{min}$	This parameter was estimated by $r_2 = \theta_{ribo} / T_{ribo}$, where θ_{ribo} is the number of ribosomes normalized by the number of proteins, and T_{ribo} is the mean protein synthesis time.
K_1	Saturation level of DNA with respect to RNAP binding	$1/\mu\text{m}^3$	This parameter was estimated from $\alpha_{RNAP} = \frac{[Z]}{K_1 + [Z]}$, where α_{RNAP} is measured in this work and $[Z]_{avg}$, the mean genome copy.
K_2	Saturation level of mRNA with respect to ribosome binding	$1/\mu\text{m}^3$	This parameter was estimated as $\alpha_{ribo} = \frac{[X]}{K_2 + [X]}$, where α_{ribo} is measured in this work and $[X]$, the mRNA concentration.
δ	mRNA degradation rate	$1/\text{min}$	$\delta = 1/\tau_L$, where τ_L is the mRNA lifetime
c	$\frac{\text{cell volume}}{\text{protein number}}$	$10^{-6}\mu\text{m}^3$	Cell volumes of different culture conditions were measured in this work.
c'	$\frac{\text{cell area}}{\text{protein number}}$	$10^{-6}\mu\text{m}^2$	Cell area of different culture conditions were measured in this work.
$[Z]_{avg}$	Average genome concentration of normal-growing cells	genome/ μm^3	Estimated by averaging the genome concentration over the cell cycle, see Appendix 1 for details.

Table 2

Initial and optimized model parameters

Parameter	Unit	Initial estimation			Optimized parameters			
		M9gly CAA	M9gly	M9ala	M9glyCAA (model A)	M9glyCAA (model B)	M9gly (model A)	M9ala (model A)
r_1	$10^{-3}/\text{min}$	1.99	0.89	0.60	1.22	1.76	0.64	0.46
r_2	$10^{-3}/\text{min}$	20.0	8.21	3.70	22.4	22.5	15.3	13.6
K_1	$1/\mu\text{m}^3$	1.40	2.99	3.03	1.17	1.55	3.02	2.78
K_2	$1/\mu\text{m}^3$	622	1807	1467	797	532	1836	1513
δ	$1/\text{min}$	0.96	0.96	0.96	1.27	1.31	1.02	0.79
c	$10^{-6}\mu\text{m}^3$	0.28	0.39	0.53	0.28	0.18	0.15	0.30
c'	$10^{-6}\mu\text{m}^2$	0.47	0.62	1.06	-	-	-	-
X_{ini}	$10^2/\text{cell}$	20.2	7.21	3.17	-	-	-	-
Y_{ini}	$10^6/\text{cell}$	3.82	1.77	0.95	-	-	-	-
$[Z]_{avg}$	genome/cell	1.40	1.99	2.02	-	-	-	-

As shown in **Figure 4A-D** (model A) and **Figure 4 – figure supplement 1** (model B), both models performed similarly after parameter optimization. While the model curves (solid lines) did not exactly match the average behavior of our experimental results (open squares), they displayed similar trends and fell within the variance of the single-cell data (dots). The models showed that multi-N cells (blue) display balanced exponential growth while the 1N cells (yellow) exhibit sub-exponential growth (**Figure 4A** and **Figure 4 – figure supplement 1A**), consistent with experiments. At the same time, both models recapitulated the observed experimental trends in active fractions of both ribosomes and RNAPs, which remained constant in multi-N cells while decaying gradually with DNA concentration in 1N cells (**Figure 4B-D** and **Figure 4 – figure supplement 1B-D**).

In the absence of published values for the RNAP kinetic constants in nutrient-poor media, we only used the simpler model A for the M9gly and M9ala conditions. Here again, the model reproduced the trends for the experimental measurements of WT cells (gray) and 1N cells (yellow) (**Figure 4 – figure supplement 2**), with one notable deviation. At cell birth, the curves of fractions of active RNAPs and ribosomes for 1N cells started slightly higher than the ones for WT cells due to a limitation of our ODE model, which uses the average concentration of genome and does not include DNA replication dynamics.

The simulation results of 1N cells suggest the following cascade of events when DNA is limiting. Lower DNA concentration results in fewer substrates for RNAPs, which reduces the transcription rate. This results in a decrease in mRNA concentration (**Figure 4E**). As mRNAs become limiting, the fraction of ribosomes engaged in translation decreases. This, in turn, decreases the rate in bulk protein synthesis, which decreases the relative growth rate. The greater the DNA dilution (through cell growth), the more severe the downstream effects become, explaining the decay in relative growth rate in 1N cells (**Figure 4E**).

Since the models predicted a reduced mRNA pool in 1N cells, we performed live-cell staining with SYTO RNASelect (Wu et al., 2020), a fluorogenic RNA-specific dye that primarily (~70% of the signal) binds mRNAs in *E. coli* based on rifampicin treatment (Bakshi et al., 2014). In our RNASelect staining experiments, we mixed 1N cells with multi-N cells of similar size ranges prior to incubation with the dye to mitigate variability in staining. We next imaged the mixed populations and distinguished 1N cells from multi-N cells by examining the difference in nucleoid number (one vs. multiple) per cell (**Figure 4F**). Single cells were then sampled to ensure that the cell area distributions of the two populations perfectly matched (**Figure 4 – figure supplement 3**). Comparison between the two sampled populations from five independent experiments revealed a reduced concentration of RNASelect signal (by ~50%) in 1N cells relative to multi-N cells for a cell size range of 4 to 10 μm^2 (**Figure 4G-H**). We also found that the RNASelect concentration ratio between 1N and multi-N cells decreased exponentially with increasing cell area and thus DNA dilution (**Figure 4I**), in line with the model prediction (**Figure 4E**). Furthermore, both the model (**Figure 4E**) and our experimental findings (**Figure 4I**) suggest that the effects of DNA limitation on the mRNA concentration already appear at physiological cell area ranges (between 2 and 3 μm^2).

To verify that the decrease in RNASelect signal in 1N cells was not caused by a global change in membrane permeability to small molecules, we performed similar staining experiments with the HaloTag dye JF549 in CRISPRi strains expressing RpoC-HaloTag (**Figure 4 – figure supplement 4A-B**). Because RpoC concentration increases with cell size in 1N cells relative to multi-N cells (**Figure 3A-B**), we expected a similar increase in the ratio of JF549 signal between these two cell types if the membrane permeability to small molecules remained unchanged. This is indeed what we observed (**Figure 4 – figure supplement 4C-E**). Hence the RNASelect results are consistent with a reduced mRNA concentration in 1N cells caused by DNA dilution and reduced global transcription activity.

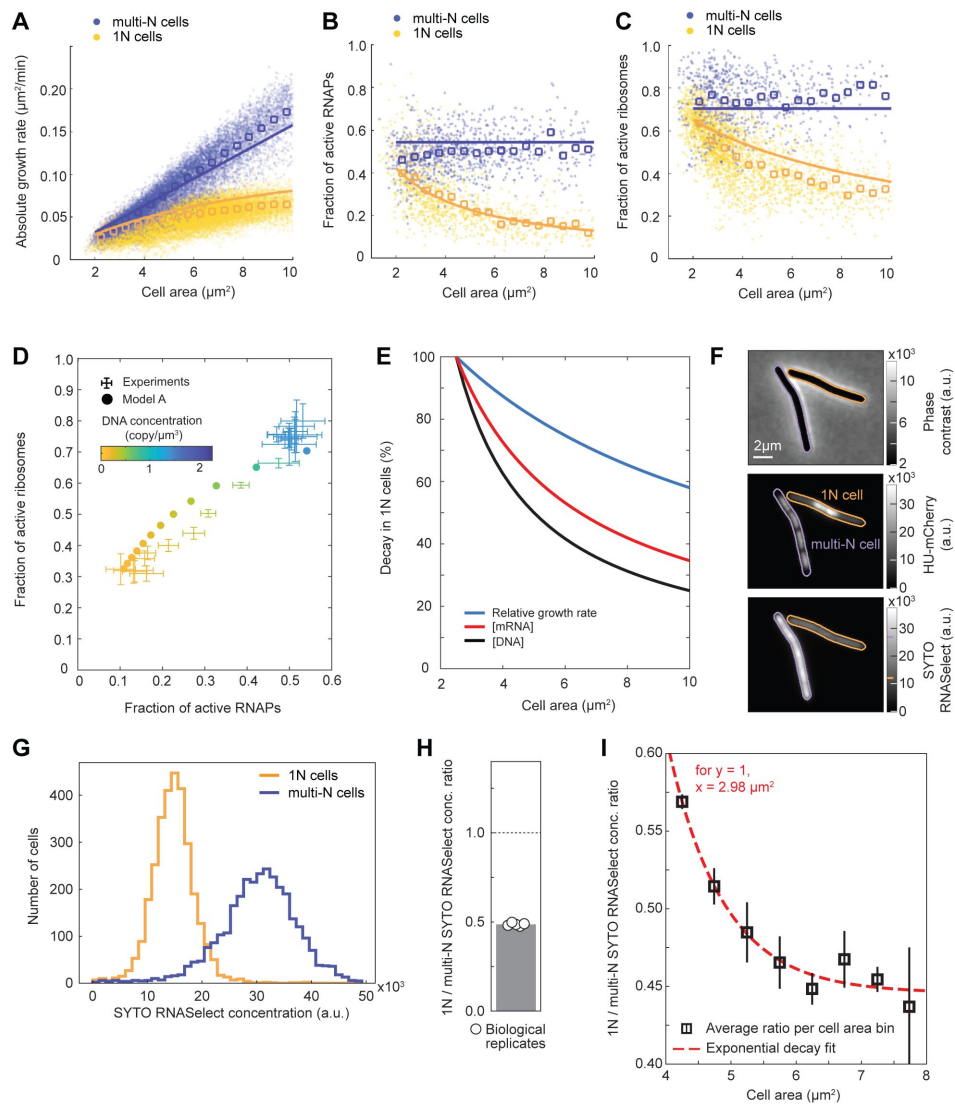


Figure 4.

Mathematical modeling and RNaselect concentration measurements in 1N and multi-N cells.

(A–C) Plots comparing simulation results of model A (solid lines) with experimental data points (dots) and averages (open squares) in the M9glyCAAT condition. The multi-N and 1N cells are indicated as blue and yellow, respectively: (A) The relation between the absolute growth rate ($\frac{dA}{dt}$) and cell area (A). (B) The relation between the active RNAP fraction and cell area. (C) The relation between the active ribosome fraction and cell area. (D) A two-dimensional diagram showing how the fractions of active RNAPs and ribosomes change with DNA concentration (colored from blue to yellow). Experimental data (with 2D error bars) from multi-N and 1N cells were combined and shown in the same plot. Simulated results (filled dots) are based on model A. (E) Plot showing the effect of DNA limitation using the ODE model A, with respect to the decay of DNA concentration, mRNA concentration and relative growth rate in 1N cells. Each quantity was normalized to their value at normal cell size (cell area = $2.5 \mu\text{m}^2$). (F) The phase contrast (top), HU-mCherry (middle) and SYTO RNaselect fluorescence (bottom) images of two representative cells from a mixed population of 1N (CRISPRi *oriC*, CJW7457) and multi-N (CRISPRi *ftsZ*, CJW7576) cells. The purple and orange ticks in the SYTO RNaselect colorbar correspond to the average fluorescence (concentration proxy) of the multi-N and the 1N cell respectively. (G) SYTO RNaselect concentration distributions for 1N and multi-N cells (3077 cells for each population from five biological replicates). (H) The average 1N/multi-N SYTO RNaselect concentration ratio (gray bar) calculated from five biological replicates (white circles). (I) RNaselect concentration ratio as a function of cell area (Mean $\pm$ SD from five biological replicates). A single exponential decay function was fitted to the average ratios ($R^2 = 97\%$). All RpoC-HaloTag-JF549 concentration comparisons or ratio calculations were performed for equal numbers of 1N and multi-N cells and overlapping cell area distributions (Figure 4 – supplement 4A).

Genome dilution incrementally changes the composition of the proteome

The fact that the relative concentrations of ribosomal proteins and RNAP subunits scaled differently with cell area in 1N cells (**Figures 2A-B** and **3A-B**) indicated that all genes are not equally impacted by DNA dilution. In yeast and mammalian cells, a decrease in the DNA-per-volume ratio has recently been demonstrated to alter the composition of the proteome, with some proteins increasing in relative concentration while others become comparatively more diluted (Lanz et al., 2023, 2022). To examine whether this effect may be conserved across domains of life, we used our proteomic TMT-MS data on the CRISPRi strains to quantify the relative concentration of each detected protein across cell areas following DNA replication or cell division arrest. For each protein, we calculated the relative change in concentration against the relative change in cell size through regression fitting, yielding a slope value. A slope of zero indicates that the relative concentration of a protein remains constant whereas a slope of -1 (or 1) means that the relative concentration is decreasing (or increasing) by two-fold with each cell size doubling (**Figure 5A**).

We found that the slope distribution was highly reproducible between biological replicates (**Figure 5 – figure supplement 1A-B**) but drastically different between 1N-rich cells and multi-N cells (**Figure 5B**, Dataset 1). In the control multi-N cells where the genome concentration does not change with cell growth, the concentration of ~94% of the detected proteins (2217/2360) remained roughly constant, with their concentrations decreasing or increasing by less than 20% per cell size doubling (i.e., slopes > -0.2 or < 0.2 ; **Figure 5B** and Dataset 1). This indicates that protein amounts largely scale with cell size, as generally assumed. However, in 1N-rich cells where the genome dilutes with cell growth, the proportion of detected proteins that maintained similarly constant concentrations across cell areas (i.e., slopes > -0.2 or < 0.2) dropped to ~37% (859/2360) (**Figure 5B** and Dataset 1). A principal component analysis on the relative protein concentration during cell growth confirmed that the relative proteome composition changes proportionally with genome dilution (1N-rich cells), whereas it remains constant when the DNA-to-cell volume is maintained (multi-N cells) (**Figure 5C**).

Interestingly, for 1N-rich cells, we also observed a slight yet significant positive correlation (Spearman $\rho = 0.23$, $p\text{-value} < 10^{-10}$, for genes within 1.35 Mb from *oriC*) between protein slope and the gene distance from *oriC* (**Figure 5D**). In other words, genes closer to *oriC* were more likely to encode proteins that dilute during DNA-limited growth (protein slope < 0) compared to distal genes (**Figure 5D**). This was irrespective of their total protein abundance based on summed ion intensities normalized by protein sequence lengths (**Figure 5E**). This may suggest the existence of a mechanism that compensates for the gene dosage associated with DNA replication.

To examine whether the proteome scaling behavior stems from changes in mRNA levels, we performed transcriptomic (RNA-seq) analysis on two biological replicates of 1N-rich cells at different time points after induction of DNA replication arrest. The two replicates were strongly correlated at the transcript level (Spearman $\rho = 0.91$, **Figure 5 – figure supplement 1C**). We also found a strong correlation in scaling behavior with cell area between mRNAs and proteins across the genome (Spearman $\rho = 0.76$, **Figure 5F** and Dataset 2), indicating that most of the changes in protein levels observed upon genome dilution take place at the mRNA level.

Discussion

In this study, we show that genome concentration acts as a rate-limiting factor for exponential cellular growth and a modulator of proteome composition in *E. coli*. These roles for genome concentration help explain why the timing of DNA replication is so robustly linked to cell volume

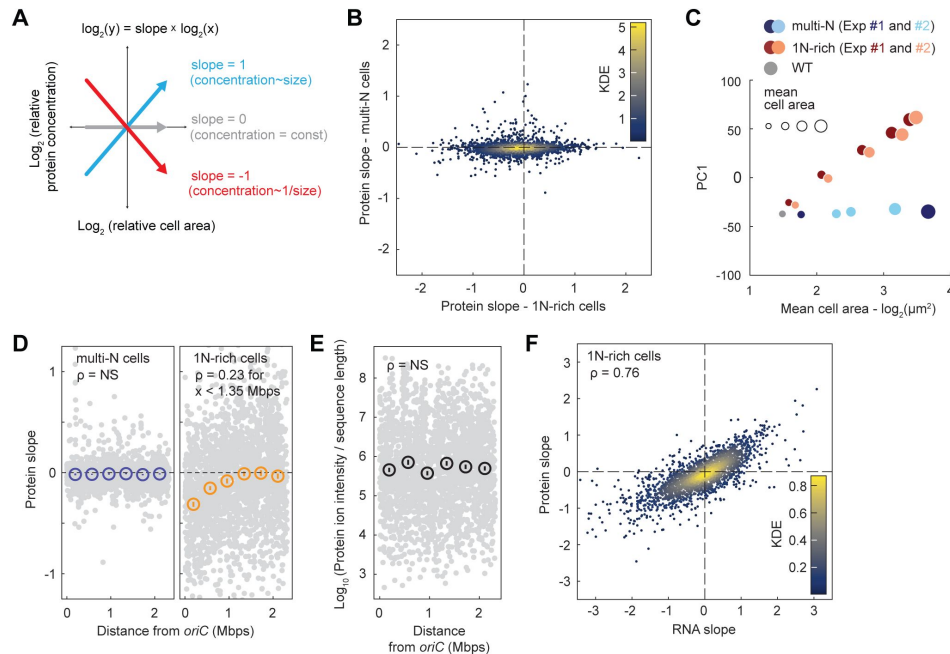


Figure 5.

Proteome and transcriptome remodeling in 1N-rich cells.

(A) Schematic explaining the calculation of the protein slopes, which describes the scaling of the relative protein concentration with cell area. **(B)** Plot showing the protein scaling (average slopes from 2 reproducible biological replicates, see also **Figure 5 – supplement 1A-B**) in 1N (x-axis) and multi-N (y-axis) cells across the proteome (2360 proteins). The colormaps correspond to a Gaussian kernel density estimation (KDE). **(C)** Plot showing the first principal component (PC1, which explains 69% of the total variance considering both 1N-rich and multi-N cells) used to reduce the dimensionality of the relative protein concentration during cell growth. The x-axis corresponds to the log-transformed cell area, whereas the marker size shows the cell area increase in linear scale. **(D)** Relationship between the absolute distance of a gene from *oriC* and the slope of the protein it encodes. Data from 2268 proteins are shown (grey markers), as well as binned data in six gene distance bins (open circles, average $\pm$ SEM, ~ 370 proteins per bin). For the multi-N cells, the Spearman correlation was not significant ($p = \text{NS}$, $p\text{-value} > 0.01$), whereas for the 1N cells, a significant Spearman correlation ($p = 0.23$, $p\text{-value} < 10^{-10}$) for genes within 1.35 Mbps of *oriC* (below the mid-point of the 4th gene distance bin). **(E)** Relationship between the distance of a gene from *oriC* and the ion intensity of its encoded protein. The protein ion intensity was divided by the protein sequence length and the quotient was log-transformed. Data from 2258 proteins are shown (grey markers) as well as binned data (open circles, average $\pm$ SEM, ~ 370 proteins per bin). The spearman correlation is not significant ($p = \text{NS}$, $p\text{-value} > 0.01$). **(F)** Correlation between average protein and mRNA slopes across the genome (2324 genes).

across environmental and genetic conditions that affect cell size (Donachie, 1968 [↗](#); Govers et al., 2024 [↗](#); Si et al., 2017 [↗](#); Zheng et al., 2016 [↗](#)).

Our data suggest that DNA limitation in 1N cells affects cell growth rate through modulation of downstream transcription and translation activities (**Figures 2 [↗](#)-4 [↗](#)** and associated figure supplements). As a result of growth-induced dilution, the genome becomes increasingly limiting as a template, decreasing the fraction of active RNAPs (**Figure 3 [↗](#)**). This indicates a sub-linear scaling of global transcription relative to the cell size. We propose that this reduced transcription activity in comparison to cells undergoing DNA replication results in a gradual decrease in total mRNA concentration (**Figure 4E-G [↗](#)**). Reduced mRNA availability would cause a decrease in the fraction of ribosomes engaged in translation (**Figure 2F [↗](#)** and **Figure 2 – supplement 4 [↗](#)**), which would, in turn, lead to sub-linear scaling of global protein synthesis and absolute growth rate with cell size (**Figure 2H [↗](#)** and **Figure 2 – supplement 4 [↗](#)**). It is also possible that the cumulative effect of small changes in proteome composition in 1N cells contributes to their growth rate defect.

The growth rate dependency on genome concentration is unlikely to be a particularity of *E. coli*, as we also observed a divergence in absolute and relative growth rates with increasing cell area between 1N and multi-N cells of *Caulobacter crescentus* (**Figure 1 – supplement 6A [↗](#)** and B). In this organism, filamenting 1N and multi-N cells were generated by depleting the DNA replication initiation factor DnaA (Gorbatyuk and Marczyński, 2001 [↗](#)) and the cell division protein FtsZ (Wang et al., 2001 [↗](#)), respectively. The 1N versus multi-N designation was confirmed by visualizing the number of chromosomal origins of replication (one versus multiple) per cell using the *parS*/ParB-eCFP labeling system (**Figure 1 – supplement 6C [↗](#)**). The fact that DNA limitation for cellular growth was also observed in *C. crescentus* is significant not only because this bacterium is distantly related to *E. coli*, but also because it has a different pattern of cell wall growth and distinct control mechanisms of DNA replication (Aaron et al., 2007 [↗](#); Banerjee et al., 2017 [↗](#); Frandi and Collier, 2019 [↗](#); Lasker et al., 2016 [↗](#); Terrana and Newton, 1975 [↗](#)). This suggests that DNA concentration may be a prevalent growth constraint across bacterial species.

We found that even a relatively small dilution in DNA concentration—as expected in DNA replication-arrested cells that are still within or close to physiological sizes—results in growth rate reduction in *E. coli* and *C. crescentus* (**Figure 1 [↗](#)** and **Figure 1 – supplements 2 [↗](#)** and **6 [↗](#)**). Thus, under our standard growth conditions, cells appear to live at the cusp of DNA limitation for cellular growth. This suggests that cells make enough—but not too much—DNA, presumably because DNA replication is a costly process that represents a significant fraction (~6% in minimal media) of the cellular energy budget (Neidhardt et al., 1990 [↗](#)). What may be the implications of living at the cusp of DNA limitation in bacteria? While *E. coli* carefully controls its DNA concentration across various conditions and growth rates at the population level (Donachie, 1968 [↗](#); Govers et al., 2024 [↗](#); Si et al., 2017 [↗](#); Zheng et al., 2016 [↗](#)), there remains some variability in DNA concentration at the single-cell level, with some cells initiating DNA replication at smaller or larger cell volumes than others (Si et al., 2019 [↗](#); Witz et al., 2019 [↗](#)). It will be interesting to explore in future studies whether this variability contributes to the known growth rate heterogeneity across isogenic cells (Lin and Jacobs-Wagner, 2022 [↗](#); Wang et al., 2010 [↗](#)). It is also tempting to speculate that changes in DNA concentration may, at least in part, contribute to the deviations from exponential growth that have been reported during the division cycle of *B. subtilis*, *E. coli*, and stalked *C. crescentus* progeny (Banerjee et al., 2017 [↗](#); Kar et al., 2021 [↗](#); Nordholt et al., 2020 [↗](#); Reshes et al., 2008 [↗](#)). More substantial forms of DNA dilution may occur under other circumstances as well. *C. crescentus* cells in freshwater lakes often form long filaments during algal blooms in the summer months (Heinrich et al., 2019 [↗](#)). These filament cells are thought to be the result of a DNA replication arrest in response to the combination of an alkaline pH, a depletion in phosphate, and an excess in ammonium (Heinrich et al., 2019 [↗](#)). Another example is illustrated by the Lyme disease agent *Borrelia burgdorferi*. This pathogen,

which forms long polyploid cells during exponential growth, experiences a progressive decrease in genome concentration (up to eightfold) in stationary phase cultures through the gradual loss of genome copies (Takacs et al., 2022 [DOI](#)).

Comparison with studies on eukaryotic cells suggests conservation of gene expression principles across domains of life. For instance, in yeast, it has been shown that the global transcription rate in G1-arrested cells is higher in diploids than haploids of similar sizes (Swaffer et al., 2023 [DOI](#)), consistent with DNA concentration being a limiting factor for transcription. Furthermore, in both yeast and mammalian cells, small G1-arrested cells display higher growth rate (or global RNA or protein synthesis rate) per cell volume than large ones that have exceeded a certain volume (Cadart et al., 2018 [DOI](#); Liu et al., 2022 [DOI](#); Neurohr et al., 2019 [DOI](#); Zatulovskiy et al., 2022 [DOI](#)). This is likely due to a change in genome concentration rather than a change in cell volume, as the relative growth rate is unaffected in very large cells as long as they undergo a proportional increase in ploidy (Mu et al., 2020 [DOI](#)).

In yeast cells, an increase in RNAP II gene occupancy combined with a decrease in mRNA turnover helps mitigate DNA dilution on global transcription activities up to a certain cell volume beyond which the compensation breaks down (Swaffer et al., 2023 [DOI](#); Zhurinsky et al., 2010 [DOI](#)). Such buffering activities, which are consistent with model predictions (**Figure 4 – figure supplement 5** [DOI](#)) (Swaffer et al., 2023 [DOI](#)), may also be at play in bacteria, potentially in a growth medium-dependent manner. For example, in our *E. coli* experiments, we noted that the global transcriptional activity (total amount of active RNAPs) strongly deviated from proportional scaling with cell size in 1N cells compared to WT and multi-N cells in all tested medium conditions (**Figure 3H** [DOI](#) and **Figure 3 – supplement 4G-H** [DOI](#)). This indicates that the genome content limits transcription. However, the global translation activity (total amount of active ribosomes) only strongly deviated from proportional scaling with cell sizes in 1N cells growing in the relatively rich medium (M9glyCAAT) condition (**Figure 2H** [DOI](#)). In poor media (M9gly and M9ala), the deviation was far less pronounced (**Figure 2H** [DOI](#) and **Figure 2 – supplement 4G-H** [DOI](#)), suggesting that a decrease in mRNA turnover and/or an increase in ribosome occupancy per transcript partially buffer(s) DNA limitation on transcription in poor media.

Another remarkable similarity between bacteria and eukaryotes is the effect of genome concentration on proteome composition. While protein abundance is typically assumed to scale with cell size in bacteria, we found that this is true at the proteome level only when ploidy also scales (**Figure 5** [DOI](#)). This requirement was also recently shown in yeast and mammalian cells (Lanz et al., 2023 [DOI](#), 2022 [DOI](#)). This conservation in scaling principle further highlights the importance of genome concentration in controlling protein expression.

At the GO term level, we did not identify any specific trends in proteome changes (Dataset 1). In eukaryotic cells, histones are known to scale in proportion with DNA rather than cell size (Claude et al., 2021 [DOI](#); Swaffer et al., 2021 [DOI](#); Wiśniewski et al., 2014 [DOI](#)). As a result, their concentration proportionally decreases (i.e., slope = -1) with growth in G1 phase. In *E. coli*, the relative abundance of some nucleoid-associated proteins (H-NS, HU and Dps) decreased with genome dilution, while others (IHF and Fis) displayed superscaling behavior (**Figure 5 – supplement 2** [DOI](#)). What determines the scaling behavior of proteins is not clear. We found that, in *E. coli*, the alteration in proteome composition largely occurs at the mRNA level (**Figure 5F** [DOI](#)) and partially involves chromosomal positioning (**Figure 5D** [DOI](#)).

Finally, given the prevalent use of *E. coli* in the biotechnological world, we hope that our findings will be helpful to future bioengineering studies and growth rate optimization efforts. We show that protein content and cellular growth depend on the ploidy-to-cell volume ratio (**Figures 1** [DOI](#) and **5** [DOI](#)). As such, models of protein expression that take into consideration the DNA concentration and the active number of RNAPs and ribosomes could provide a starting point to identify the parameter space that leads to growth rate improvement.

Materials and methods

Bacterial strains and growth conditions

Bacterial strains and their construction can be found in [Table 3](#). Oligomers used for polymerase chain reaction (PCR) are listed in [Table 4](#). Transductions and Gibson assemblies were performed as described previously ([Ely, 1991](#); [Thomason et al., 2007](#)).

E. coli strains were grown at 37°C in M9 minimal media with different supplements: 0.2% glycerol, 0.1% casamino acids and 1 µg/ml thiamine (M9glyCAAT); 0.2% glycerol (M9gly); or 0.2% L-alanine (M9ala). For microscopy, cells were grown in culture tubes to stationary phase, diluted 10,000-fold, and grown until they reached an optical density (OD₆₀₀) between 0.05 and 0.2. For imaging, cells were then spotted onto a 1% agarose pad on a glass slide prepared with the appropriate M9 medium and covered by a #1.5 thickness coverslip.

For time-lapse microscopy experiments with the CRISPRi strains ([Li et al., 2016](#)), the cells were induced by adding L-arabinose (0.2%) to liquid cultures after which a sample (~1 µl) was immediately collected and spotted on an agarose pad containing the appropriate growth medium supplemented with 0.2% L-arabinose. This was promptly followed by imaging of individual cells. To determine the 1N status of CRISPRi *oriC* cells, we monitored the last division and number of nucleoids based on HU-mCherry fluorescence. In [Figure 1B](#), the time point “0 min” refers to the time when cells have reached the 1N status.

For population microscopy experiments with the CRISPRi strains, the cells were induced with 0.2% L-arabinose and allowed to grow in normal conditions until a specific time point (depending on the growth medium and strain) was reached, after which the cells were spotted on an agarose pad containing the appropriate growth medium and 0.2% L-arabinose, and imaged (see information for each experiment in the corresponding Figure). Note that the CRISPRi strains do not metabolize arabinose due to the *araBAD* deletion.

For the TMT-MS experiments, CRISPRi *oriC* (SJ_XTL676) cells were supplemented with 0.2% L-arabinose and allowed to grow in liquid M9glyCAAT cultures at 37°C for 0, 120, 180, 240 and 300 min before harvesting, while CRISPRi *ftsZ* (SJ_XTL229) cells were collected after 0, 60, and 120 min after arabinose addition. For RNAseq experiments, CRISPRi *oriC* (SJ_XTL676) cells were supplemented with 0.2% L-arabinose and allowed to grow liquid M9glyCAAT cultures 37°C for 60, 120 200, and 240 min before harvesting.

Strains carrying the *dnaC2* mutation were grown at a permissive temperature of 30°C and then shifted to 37°C to block replication initiation. Transcription inhibition and mRNA depletion were achieved by exposing cells to 200 µg/ml rifampicin for 30 min before spotting cells onto a 1% agarose pad containing the appropriate M9 medium and rifampicin concentration. The HaloTag was labeled with Janelia Fluor 549 (JF549) ligand ([Grimm et al., 2015](#)) as described previously ([Banaz et al., 2019](#)). Briefly, cells were incubated with 2.5 µM of the JF549 ligand for 30 min while shaking, washed 5 times with growth medium, and allowed to recover for several generations (while remaining in exponential phase) prior to imaging.

C. crescentus strains were grown at 22°C in PYE (2 g/L bacto-peptone, 1 g/L yeast extract, 1 mM MgSO₄, 0.5 mM CaCl₂) supplemented with 0.03% xylose to induce production of the essential protein FtsZ ([Wang et al., 2001](#)) or DnaA ([Gorbatyuk and Marczyński, 2001](#)). For microscopy, cells were grown in overnight cultures, then diluted at least 1:10,000 in fresh medium and grown to exponential phase (OD₆₆₀ nm < 0.3). To induce cell filamentation by depleting FtsZ or DnaA, xylose was removed from the medium by pelleting cells, washing twice with PYE, and

Table 3.

Strains used in this study

. The abbreviations *kan*, *cat*, and *spec* refer to gene cassette insertions conferring resistance to kanamycin, chloramphenicol, and spectinomycin, respectively. These insertions are flanked by Flp site-specific recombination sites (*frt*) that allow the removal of the insertion using Flp recombinase from plasmid pCP20 (Cherepanov and Wackernagel, 1995 [\[1\]](#)).

Identifier	Genotype	Source
CJW2591	CB15N $\Delta bla6$ <i>ftsZ</i> ::pBGent-pXyl- <i>ftsZ</i>	BamHI/HindIII fragment was cut from YF1585 and cloned into pBGent. The
		resulting plasmid was integrated into LS107 (Wang et al., 2001).
CJW3673	CB15N <i>ftsZ</i> ::pBGentpXyl- <i>ftsZ</i> <i>parB</i> :: <i>ecfp-parB</i>	UV-inactivated ϕ Cr30 phage lysate was prepared from CJW2591 and transduced into MT190.
CJW4823	CB15N <i>parB</i> :: <i>ecfp-parB</i> <i>dnaA</i> :: Ω <i>xylX</i> ::pGM2195	UV-inactivated ϕ Cr30 phage lysate was prepared from GM2471. First, pGM2195 was transduced into MT190, followed by a second transduction of Ω into the resulting strain to disrupt the endogenous <i>dnaA</i> gene.
CJW6737	DH5 α /pKFSS1_ <i>parS</i> ^{PMT1}	<i>parS</i> ^{PMT1} was amplified from FH3518 (Nielsen et al., 2006) using oligonucleotides NT157 and NT158, digested with BamHI and KpnI, and cloned into the BamHI/KpnI sites of pKFSS1_2 (Takacs et al., 2018).
CJW7019	MG1655 <i>rpsB</i> :: <i>rpsB-msfgfp-kan</i>	(Gray et al., 2019)
CJW7079	CB15N <i>rplA(L1)</i> :: <i>rplA(L1)-mCherry</i>	CJW7080 was subjected to counterselection on 0.3% xylose.
CJW7080	CB15N <i>rplA(L1)</i> ::pNPTS138- <i>rplA(L1)-mCherry-rplA(L1)DOWN</i>	The vector pNPTS138 (M. R. Alley, unpublished) was PCR-amplified using oligomers JSG_018 and JSG_019. A ~500-bp fragment of the 3' end of the <i>rplA(L1)</i> gene, and a ~500 bp fragment immediately downstream of the <i>rplA</i> gene (<i>rplA(L1)DOWN</i>), were amplified from a CB15N genome template using oligomers JSG_024 and JSG_025, and JSG_020 and JSG_021, respectively. <i>mCherry</i> was amplified from MTL54218 (Thanbichler et al., 2007) and a linker was introduced using JSG_022 and JSG_023. All products were assembled into a plasmid via Gibson assembly. The resulting plasmid, which was verified by sequencing junctions, was transformed into S17-1 and introduced into CB15N by conjugation with selection for

		kanamycin and nalidixic acid resistance.
CJW7339	MG1655 <i>hupA::hupA-mCherry-kan</i>	<i>hupA::hupA-mCherry-kan</i> from CJW5556 (Paintdakhi et al., 2016) was moved into MG1655 by P1 transduction.
CJW7374	MG1655 <i>hupA::hupA-mCherry dnaC::dnaC2(ts) mdoB::kan</i>	The kanamycin resistance cassette was excised from CJW7339 by expressing the Flp site-specific recombinase from pCP20 (Cherepanov and Wackernagel, 1995). <i>dnaC::dnaC2(ts) mdoB::kan</i> from FW1957 (Wu et al. 2010) was subsequently moved into this strain by P1 transduction.
CJW7411	MG1655 <i>glmS1868::parS(pMT1)-frt-kanR-frt</i>	Plasmid backbone was amplified from CJW6919 (Kim et al., 2019) using oligonucleotides AP0039 and AP0040, pMT1 <i>parS</i> was amplified from CJW6737 using AP0041 and AP0042, and the fragments were ligated by Gibson assembly. <i>parS(pMT1)-frt-kanR-frt</i> was amplified from the resulting plasmid with AP0078 and AP0079. The amplicon was inserted between <i>glmS</i> and <i>pstS</i> by λ -red mediated recombination (Datsenko and Wanner, 2000).
CJW7457	MG1655 <i>lacY(A177C) araFGH::spec ΔlacI ΔaraE araBAD::dCas9 galM<PBBa-J23119-sgRNA(oriC)-(S. pyogenes terminator)-(rrnB terminator)>gmpA hupA::hupA-mCherry-kan</i>	<i>hupA::hupA-mCherry-kan</i> from CJW5556 (Paintdakhi et al., 2016) was moved into SJ_XTL676 by P1 transduction.
CJW7473	CB15N <i>rplA(L1)::rplA(L1)-mCherry ftsZ::pBGentpXyl-ftsZ</i>	UV-inactivated ϕ Cr30 phage lysate was prepared from CJW7080 and used for transduction into CJW3673. Transductants were selected on kanamycin, and then counterselected on 0.3 % xylose.
CJW7477	MG1655 <i>lacY(A177C) araFGH::spec ΔlacI ΔaraE</i>	The kanamycin resistance cassette was excised from CJW7457 by

Table 3. (continued)

	<i>araBAD::dCas9 galM<PBa- J23119-sgRNA(oriC)-(S. pyogenes terminator)-(rrnB terminator)>gmpA hupA::hupA- mCherry rpoC::rpoC-ygfp-kan</i>	expressing the Flp site-specific recombinase from pCP20 (Cherepanov and Wackernagel, 1995). <i>rpoC::rpoC-ygfp-kan</i> from RL7470 was subsequently moved into this strain by P1 transduction.
CJW7478	MG1655 <i>lacY(A177C)</i> <i>araFGH::spec ΔlacI ΔaraE</i> <i>araBAD::dCas9 galM<PBa- J23119-sgRNA(oriC)-(S. pyogenes terminator)-(rrnB terminator)>gmpA hupA::hupA- mCherry rpsB::rpsB-msfgfp-kan</i>	The kanamycin resistance cassette was excised from CJW7457 by expressing the Flp site-specific recombinase from pCP20 (Cherepanov and Wackernagel, 1995). <i>rpsB::rpsB-msfgfp-kan</i> from CJW7019 was subsequently moved into this strain by P1 transduction.
CJW7500	CB15N <i>rplA(L1)::rplA(L1)- mCherry dnaA::Ω xylX::pGM2195</i>	UV-inactivated φCr30 phage lysate was prepared from GM2471. A double transduction was performed into CJW7079.
CJW7517	MG1655 <i>lacY(A177C)</i> <i>araFGH::spec ΔlacI ΔaraE</i> <i>araBAD::dCas9 galM<PBa- J23119-sgRNA(oriC)-(S. pyogenes terminator)-(rrnB terminator)>gmpA gtrA::P58- mCherry-parB(pMT1)</i> <i>parS(pMT1) at ori1 hupB::hupB- cfp-kan</i>	The strain was generated by first moving <i>gtrA::P58-mCherry-parB(pMT1)-cat</i> (Wiktor et al., 2021) into SJ_XTL676 by P1 transduction. Next, <i>parS(pMT1)-kan</i> at <i>ori1</i> from CJW7411 was moved into the strain by P1 transduction. The resistance genes were excised by expressing the Flp site-specific recombinase from pCP20 (Cherepanov and Wackernagel, 1995). Finally, <i>hupB::hupB-cfp-kan</i> from CJW4656 (Gray et al., 2019) was moved into the strain by P1 transduction.
CJW7518	MG1655 <i>lacY(A177C)</i> <i>araFGH::spec ΔlacI ΔaraE</i> <i>araBAD::dCas9 galM<PBa- J23119-sgRNA(oriC)-(S. pyogenes terminator)-(rrnB terminator)>gmpA hupA::hupA- mcherry ΔrelA251::kan</i> <i>ΔspoT207::cat</i>	The kanamycin resistance cassette was excised from CJW7457 by expressing the Flp site-specific recombinase from pCP20 (Cherepanov and Wackernagel, 1995). <i>ΔrelA251::kan</i> and <i>ΔspoT207::cat</i> were subsequently moved into this strain by two P1 transductions from strain CF1693 (Xiao et al., 1991).

Table 3. (continued)

CJW7519	MG1655 <i>rpoC::rpoC-HaloTag-kan</i>	The strain was generated by first creating <i>rpoC-HaloTag-kan</i> amplicon with oligonucleotides JSG_148 and JSG_149 from strain JM41 (Mäkelä and Sherratt, 2020). This amplicon was used to replace the endogenous <i>rpoC</i> by λ -red mediated recombination (Datsenko and Wanner, 2000). Finally, <i>rpoC::rpoC-HaloTag-kan</i> was moved into MG1655 by P1 transduction.
CJW7520	MG1655 <i>lacY(A177C) araFGH::spec ΔlacI ΔaraE araBAD::dCas9 galM<PBBa-J23119-sgRNA(oriC)-(S. pyogenes terminator)-(rrnB terminator)>gmpA rpoC::rpoC-HaloTag-kan</i>	<i>rpoC::rpoC-HaloTag-kan</i> from CJW7519 was moved into SJ_XTL676 by P1 transduction.
CJW7522	MG1655 <i>lacY(A177C) araFGH::spec ΔlacI ΔaraE araBAD::dCas9 galM<PBBa-J23119-sgRNA(oriC)-(S. pyogenes terminator)-(rrnB terminator)>gmpA hupA::hupA-mCherry ΔrecA::kan</i>	<i>recA</i> was replaced by <i>kan</i> using oligonucleotides JSG_173 and JSG_174 in MG1655 by λ -red mediated recombination (Datsenko and Wanner, 2000). The kanamycin resistance cassette was excised from CJW7457 by expressing the Flp site-specific recombinase from pCP20 (Cherepanov and Wackernagel, 1995) and <i>ΔrecA::kan</i> was subsequently moved into the strain by P1 transduction.
CJW7527	MG1655 <i>lacY(A177C) araFGH::spec ΔlacI ΔaraE araBAD::dCas9 galM<PBBa-J23119-sgRNA(ftsZ)-(S. pyogenes terminator)-(rrnB terminator)>gmpA rpoC::rpoC-HaloTag-kan</i>	<i>rpoC::rpoC-HaloTag-kan</i> from CJW7519 was moved into SJ_XTL229 by P1 transduction.
CJW7528	MG1655 <i>rpsB::rpsB-HaloTag-kan</i>	The strain was generated by first creating <i>rpsB-HaloTag-kan</i> amplicon with oligonucleotides JSG_157 and JSG_158 from strain JM41 (Mäkelä and Sherratt, 2020). This amplicon was used to replace the endogenous <i>rpsB</i> by λ -red mediated

Table 3. (continued)

		recombination (Datsenko and Wanner, 2000). Finally, <i>rpsB::rpsB-HaloTag-kan</i> was moved into MG1655 by P1 transduction.
CJW7529	MG1655 <i>lacY(A177C) araFGH::spec ΔlacI ΔaraE araBAD::dCas9 galM<PBba-J23119-sgRNA(oriC)-(S. pyogenes terminator)-(rrnB terminator)>gmpA rpsB::rpsB-HaloTag-kan</i>	<i>rpsB::rpsB-HaloTag-kan</i> from CJW7528 was moved into SJ_XTL676 by P1 transduction.
CJW7530	MG1655 <i>lacY(A177C) araFGH::spec ΔlacI ΔaraE araBAD::dCas9 galM<PBba-J23119-sgRNA(ftsZ)-(S. pyogenes terminator)-(rrnB terminator)>gmpA rpsB::rpsB-HaloTag-kan</i>	<i>rpsB::rpsB-HaloTag-kan</i> from CJW7528 was moved into SJ_XTL229 by P1 transduction.
CJW7531	CB15N <i>rpoC::mCherry-rpoC</i>	Two DNA segments of vector pNPTS138 (M. R. Alley, unpublished) were PCR-amplified using oligomers JSG_134 and JSG_137, and JSG_135 and JSG_136. A ~900 bp fragment upstream of the <i>rpoC</i> gene and ~900 bp of the 5' end of <i>rpoC</i> were amplified from a CB15N genome template using oligomers JSG_128 and JSG_129, and JSG_132 and JSG_133, respectively. <i>mCherry</i> was amplified from MTL#4218 (Thanbichler et al., 2007) and a linker was introduced using oligomers JSG_130 and JSG_131. All products were assembled into a plasmid via Gibson assembly. Proper plasmid assembly was verified by sequencing junctions. Plasmid was transformed into S17-1 and introduced into CB15N by conjugation with selection for kanamycin and nalidixic acid resistance. Second homologous recombination was achieved by counterselection on 0.3 % xylose.

Table 3. (continued)

Table 3. (continued)

CJW7535	CB15N <i>rpoC::mCherry-rpoC dnaA::Ω xylX::pGM2195</i>	UV-inactivated φCr30 phage lysate was prepared from GM2471. First, pGM2195 was transduced into CJW7531, followed by a second transduction of Ω into the resulting strain to disrupt the endogenous <i>dnaA</i> gene.
CJW7563	MG1655 <i>lacY(A177C) araFGH::spec ΔlacI ΔaraE araBAD::dCas9 galM<PBBa-J23119-sgRNA(ftsZ)-(S. pyogenes terminator)-(rrnB terminator)>gmpA rpoC::rpoC-ygfp-kan</i>	<i>rpoC::rpoC-ygfp-kan</i> from RLG7470 was moved into SJ_XTL229 by P1 transduction.
CJW7564	MG1655 <i>lacY(A177C) araFGH::spec ΔlacI ΔaraE araBAD::dCas9 galM<PBBa-J23119-sgRNA(ftsZ)-(S. pyogenes terminator)-(rrnB terminator)>gmpA rpsB::rpsB-msfgfp-kan</i>	<i>rpsB::rpsB-msfgfp-kan</i> from CJW7019 was moved into SJ_XTL229 by P1 transduction.
CJW7569	CB15N <i>rpoC::mCherry-rpoC ftsZ::pBGentpXyl-ftsZ</i>	UV-inactivated φCr30 phage lysate was prepared from CJW2591 and transduced into CJW7531.
CJW7576	MG1655 <i>lacY(A177C) araFGH::spec ΔlacI ΔaraE araBAD::dCas9 galM<PBBa-J23119-sgRNA(ftsZ)-(S. pyogenes terminator)-(rrnB terminator)>gmpA hupA::hupA-mCherry-kan</i>	<i>hupA::hupA-mCherry-kan</i> from CJW5556 (Paintdakhi et al., 2016) was moved into SJ_XTL229 by P1 transduction.
GM2471	CB15N <i>dnaA::Ω xylX::pGM2195</i>	(Gorbatyuk and Marczynski, 2001)
LS107	CB15N <i>Δbla6</i>	(West et al., 2002)
MG1655	<i>F- lambda- ilvG- rfb-50 rph-1</i>	(Guyer et al., 1981)
MT190	CB15N <i>parB::ecfp-parB</i>	(Thanbichler and Shapiro, 2006)
RLG7470	<i>rpoC::rpoC-ygfp-kan</i>	(Bakshi et al., 2012)
SJ_XTL229	<i>lacY(A177C) araFGH::spec ΔlacI ΔaraE araBAD::dCas9 galM<PBBa-J23119-sgRNA(ftsZ)-(S. pyogenes terminator)-(rrnB terminator)>gmpA</i>	(Li et al., 2016)
SJ_XTL676	<i>lacY(A177C) araFGH::spec ΔlacI ΔaraE araBAD::dCas9 galM<PBBa-J23119-sgRNA(oriC)-(S. pyogenes terminator)-(rrnB terminator)>gmpA</i>	(Si et al., 2017)
YB1585	CB15N <i>ftsZ::pBGST18-pXyl-ftsZ</i>	(Wang et al., 2001)

Table 4.

Oligonucleotides used in this study.

Name	Sequence
AP0039	ACCCAACCTTAATCGCTGTAGGCTGGAGCTGCTTCG
AP0040	ATTAATTGCGTTGCGCAGTTCCTACTCTCGCATGGG
AP0041	GAGAGTAGGGAAGTGCACAACGAATTAATGTGAGTTAGC
AP0042	CAGCTCCAGCCTACAGCGATTAAGTTGGGTAACGCCA
AP0078	ATCAAACATCCTGCCAACTCCATGTGACAAACCGTCATCTTCGGCTACTTGCAACGCAA TTAATGTGAGTTAGC
AP0079	ACAAACATTAATAACGAAGAGATGACAGAAAAATTTTCATTCTGTGACAGCCCTGATTC CGTGAGGATGC
JSG_018	GCCCGAGGGCACGTGACGCGTCTCCGGATG
JSG_019	ACCTCGGCGATCGCGAACGCCGAAGCTAGCGAATTCGTGGATC
JSG_020	GAATTCGCTAGCTTCGGCGTTCGCGATCGCCGAGGTG
JSG_021	GGCATGGACGAGCTGTACAAGTAAGCCACCAAGTTTTAGTAAGCG
JSG_022	CGCTTACTGGGTACGCTTACTAAACTTGGGTGGCTTACTTGTACAGCTCGTCCATGC C
JSG_023	AACGTACGCGTCACCGGTGCGCCACCATGGTGAGCAAGGGCGAG
JSG_024	GGCCGACCGGTGACGCGTAACGTTTGAATTGGCGCCGATCGAGCTGATG
JSG_025	CGGAGACGCGTCACGTGCCCTCGGGCACCGGCC
JSG_128	TCCGTAATACGACTCACTTAAGGCCTTGACGATGGTCAAGGTCTTCGTGG
JSG_129	GCCATGTTATCCTCCTCGCCCTTGCTACCATCTTGTTTCTTCTGCGGG
JSG_130	TTTTCGCGGGAACCCGCGAGAAGGAACCAAGATGGTGAGCAAGGGCGAG
JSG_131	TTCAGGACTTCTGGTTCATGCTGCCGCTGCCGCTGCCCTTGACAGCTCGTCCATGC
JSG_132	AGCTGTACAAGGGCAGCGGCAGCGCAGCATGAACCAGGAAGTCCTGAAC
JSG_133	ATGTACAGGCATGCGTCGACCCTTAAGCATCCGCTTTTCGTTG
JSG_134	CGCAACGAAAAGCGGATGCTTAGAGGGTCGACGCATGC
JSG_135	CACGAAGACCTTGACCATCGTCAAGGCCTTAAGTGAGTCG
JSG_136	TTATCCGTACTCCTGATGATGC
JSG_137	GGGATCGCAGTGGTGAGTAAC
JSG_148	GCCTGGCAGAACTGCTGAACGCAGGTCTGGGCGGTTCTGATAACGAGTAATCGGCTG GCTCCGCTGCT
JSG_149	CATAAAAAAACCCGCGAAGCGGGTTTTACGTTATTTGCGGATTAACGAGGATCCCA TATGAATATCCTCCTT
JSG_157	GTTCTCAGGATCTGGCTTCCAGGCGGAAGAAAGCTTCGTAGAAGCTGAGTCGGCTGG CTCCGCTGCT
JSG_158	TTGCCGCTTTCTGCAACTCGAACTATTTTGGGGGAGTTATCAAGCCTTAGGATCCCAT ATGAATATCCTCCTT
JSG_173	AAAAAAGCAAAAAGGGCCGAGATGCGACCCTTGTGTATCAACAAGACGAGTGTAGG CTGGAGCTGCTTC
JSG_174	CAACAGAACATATTGACTATCCGGTATTACCGGCATGACAGGAGTAAATGTACAAG AAAGCTGGGTACG
NT157	TATGGATCCTGTTTTTACCACGCCAATTCATGG
NT158	TATGGTACCGAGGTTGAAAAGCGTGGTG

resuspending in fresh PYE. Cultures were then allowed to grow for an additional 30 min to allow ongoing cell division cycles to complete before spotting on 1% agarose pads containing PYE but lacking xylose to deplete FtsZ or DnaA. To estimate the concentration of fluorescently-labeled ribosomes and RNAPs by fluorescence microscopy, cultures were sampled at 0, 4, 8, and sometimes 12 h following xylose depletion with 30 min outgrowth to allow late predivisional cells time to divide before spotting. For timelapse microscopy, new pads were spotted with cells from the original culture at 0, 4, 8, and sometimes 12 h following xylose removal with 30 min outgrowth in a liquid PYE medium.

Epifluorescence microscopy

For *E. coli*, phase contrast and fluorescence imaging (except for the RpoC-HaloTag-JF549 epifluorescence experiment), were performed on a Nikon Ti2 microscope equipped with a Perfect Focus System, a 100x Plan Apo λ 1.45 NA oil immersion objective, a motorized stage, a Prime BSI sCMOS camera (Photometrics), and a temperature chamber (Okolabs). Fluorescence emission was collected during a 100 or 200 ms exposure time provided by a Spectra III Light Engine LED excitation source (Lumencor): mCherry – 594 nm excitation, DAPI/FITC/TxRed filter cube (polychroic FF-409/493/596-Di02, triple-pass emitter FF-1-432/523/702-25); GFP and SYTO RNaselect – 488 nm excitation, DAPI/FITC/TxRed polychroic filter cube, and an ET525/50M emission filter; YFP – 514 nm excitation, CFP/YFP/mCherry filter cube (polychroic FF-459/526/596-Di01, triple-pass emitter FF-1-475/543/702-25), and a FF02-525/40-25 emission filter. The microscope was controlled using NIS-Elements AR. For time-lapse imaging, phase images were collected every 5 min.

Epifluorescence snapshots of RpoC-HaloTag-JF549 were taken using a Nikon Ti microscope, equipped with a Perfect Focus System, a 100x Plan Apo λ 1.45 NA oil immersion objective, a motorized stage, an ORCA Flash 4.0 camera (Hamamatsu) and a temperature chamber (Okolabs). Fluorescence emission was collected during a 200 ms exposure time provided by a Sola solid state white light source (Lumencor) and a Cy3 filter cube (excitation AT545/25x, dichroic T565lpxr, emission ET605/70m). The microscope was controlled using NIS-Elements AR.

For *C. crescentus*, phase contrast and fluorescence imaging were performed on a Nikon Ti-E microscope equipped with a Perfect Focus System, a 100x Plan Apo λ 1.45 NA oil immersion objective, a motorized stage, an Orca-Flash4.0 V2 142 CMOS camera (Hamamatsu) at room temperature. Chroma filter sets were used to acquire fluorescence images: CFP (excitation ET436/20x, dichroic T455lp, emission ET480/40m) and mCherry (excitation ET560/40x, dichroic T585lpxr, emission ET630/75m). The microscope was controlled using NIS-Elements AR. For time-lapse imaging, phase images were collected every 2.5 min.

Photoactivated localization microscopy

For single-molecule photoactivated localization (PALM) microscopy, coverslips were plasma-cleaned of background fluorescent particles using a plasma cleaner (PDC-32G, Harrick Plasma). Live cell PALM microscopy was performed on a Nikon N-STORM microscope equipped with a Perfect Focus System and a motorized stage. JF549 fluorescence was measured using an iXon3 DU 897 EMCCD camera (Andor) and excited from a 50 mW 561 nm laser (MLC400B laser unit, Agilent) with 50% transmission. The laser was focused through a 100x Apo TIRF 1.49 NA oil immersion objective (Nikon) onto the sample using an angle for highly inclined thin illumination to reduce background fluorescence (Tokunaga et al., 2008 [\[1\]](#)). Fluorescence emission was filtered by a C-N Storm 405/488/561/647 laser quad set. Transmission illumination was used to gather brightfield images. PALM movies of 20,000 frames were acquired with continuous laser illumination and a camera frame rate of 10.7 ms.

SYTO RNASelect staining experiments

In order to compare the mRNA concentration between 1N and multi-N cells, exponentially growing CJW7576 (CRISPRi *ftsZ*) and CJW7457 (CRISPRi *oriC*) cells were stained with the fluorogenic SYTO RNASelect dye (InvitrogenTM, S7576) after CRISPRi induction with 0.2% L-arabinose. To ensure overlapping cell area distributions in the absence of cell division, considering the measured growth rate differences between the 1N and multi-N cells, CRISPRi *oriC* was induced for 3.5 h whereas CRISPRi *ftsZ* was induced for 1.5 h. Then, the two populations were mixed at equal optical densities (OD_{600}) and stained with 0.5 μ M SYTO RNASelect for 15 min at 37°C with shaking. For each staining, a fresh 5 μ M SYTO RNASelect stock was prepared in L-arabinose-containing medium. Stained cells (~0.5 μ L) were spotted on a 1% agarose pad prepared with the same growth medium with L-arabinose for imaging. Five biological replicates were performed.

RpoC-HaloTag-JF549 staining experiments for epifluorescence snapshots

For **Figure 4 – supplement 5** [↗](#), CRISPRi *oriC* in CJW7520 cells was induced for 3.5 h whereas CRISPRi *ftsZ* (CJW7527) was induced for 1.5 h with 0.2% L-arabinose to obtain a similar range of cell sizes for imaging. Then the two populations were mixed at equal optical densities (OD_{600}) and stained with 2.5 μ M JF549 at 37°C for 30 min with shaking. The mixed cells were then washed three times with L-arabinose-containing (0.2%) medium. All washes were performed at 4°C, using ice-cold medium to block cell growth and avoid the dilution of the dye. Then, the cells (~0.5 μ L) were spotted on a 1% agarose pad prepared with the same L-arabinose-containing growth medium for imaging. Two biological replicates were performed.

Image processing and data analysis

Data analysis was done in MATLAB (Mathworks) (MAIN_pop_analysis.m script for epifluorescence snapshots, MAIN_timelapse.m script for timelapse movies and MAIN_mol_tracking.m script for single-molecule tracking experiments), except for segmentation of phase contrast images, which was done in Python (segmentationRun.py) using a convolutional neural network, the Nested-Unet (Wiktor et al., 2021 [↗](#); Zhou et al., 2020 [↗](#)), and the analysis of the SYTO RNASelect and RpoC-HaloTag-JF549 epifluorescence (snapshots_analysis_UNET_version_ND2.py class) that was also performed in Python 3.9 (DNA_limitation_python_environment.yml). The Nested-Unet network was trained for our microscopy setup using PyTorch 1.7.0 and NumPY 1.19.2 (trainerWrapper.py) (Harris et al., 2020 [↗](#); Paszke et al., 2019 [↗](#)).

Cell area masks from segmentation were linked between time-lapse frames based on maximum overlap (trackCells.m). Two masks linked to the same cell area were considered a cell division event. To estimate the absolute growth rate, the cell area over time was smoothed by a sliding-average window of five data points (time interval 5 min) and the difference in the cell area between consecutive frames was calculated. The relative growth rate was calculated by dividing the absolute growth rate by the cell area. Cell areas were converted into volumes using Oufiti's scripts (Paintdakhi et al., 2016 [↗](#)). To avoid bias from cells reaching sizes too large to support growth, the time-lapse data for filamenting cells was truncated based on their maximum absolute growth rate. We confirmed that the relative reduction in the growth rate of 1N in comparison to multi-N cells was not due to longer time-lapse duration as we observed no differences when acquiring shorter 2 h time-lapse sequences (**Figure 1 – figure supplement 5A** [↗](#)). Finally, we confirmed that cells were under stable growth conditions prior to starting imaging by showing that relative growth remained constant from the beginning of acquisition (**Figure 1 – figure supplement 5B** [↗](#)).

Fluorescent ParB-mCherry spots were detected by fitting a 2D Gaussian function to raw image data (detectSpots.m). First, the fluorescent image was filtered using a bandpass filter to identify the local maxima. Next, the local maxima were fitted by a Gaussian function and a spot quality score was calculated based on spot intensity and quality of a Gaussian fit ($\text{Intensity} * \text{quality}_{\text{fit}} / \sigma_{\text{fit}}$). The spot score threshold was determined by visual inspection of the training data and was set to remove poor-quality spots from analysis.

The number of HU-mCherry-labeled nucleoid areas were detected using Otsu's thresholding (multithresh.m) (Otsu, 1979 [\[link\]](#)). Minimum and maximum area thresholds for an individual nucleoid were determined by measuring the number of fluorescent spots of ParB protein fusion in HU-labeled nucleoid areas of a strain (CJW7517) carrying a *parS* site from plasmid pMT1 at *ori1* (Figure 1 – supplement 3A [\[link\]](#)). Only cells containing a single nucleoid were considered 1N cells. To measure the total fluorescence intensity of a cell, the median intensity of the area outside the cell areas was subtracted from the fluorescent intensity of each pixel of a cell, and the intensity of all pixels was summed together.

For the SYTO RNaselect and RpoC-HaloTag-JF549 epi-fluorescence snapshot experiments (Figure 4F-I [\[link\]](#), Figure 4 – supplement 4 [\[link\]](#) and Figure 4 – supplement 5 [\[link\]](#)), the nucleoid objects were segmented using the `segment_nucleoids` function in the `snapshots_analysis_UNET_version_ND2.py` class. This function combines a Laplacian of Gaussian (LoG) filter, an adaptive filter, and a hard threshold to detect the nucleoid boundaries and distinguish between 1N and multi-N cells in the mixed populations. The image filters were applied using the scikit-image (Van Der Walt et al., 2014) and NumPy (Harris et al., 2020 [\[link\]](#)) Python libraries. For the SYTO RNaselect-stained cells, the HU-mCherry fluorescence was used to segment the nucleoid objects. For the RpoC-HaloTag-JF549-stained cells, the fluorescence of RpoC-HaloTag-JF549 bound to nucleoids was used to segment the nucleoid objects.

The SYTO RNaselect concentration corresponds to the total fluorescence of the fluorogenic dye within the cell boundaries of the cell mask, divided by the area of the cell mask. Similarly, the RpoC-HaloTag-JF549 concentration was calculated in arbitrary units.

To ensure that the SYTO RNaselect or the RpoC-HaloTag-JF549 fluorescence was compared for the same distributions of cell areas between the 1N and multi-N cells in the mixed populations, random sampling was performed in each cell area bin using a sample size equal to the smaller cell number between the 1N and multi-N cells (Figure 4 – supplements 4 [\[link\]](#) and 5B [\[link\]](#)). For example, if in each cell area bin, there were 200 1N cells and 100 multi-N cells, 100 cells were randomly sampled from the 1N population to match the sample size of the multi-N population. Bins with less than 50 cells per population across biological replicates were removed from the analysis. This sampling was performed one time to compare the distributions of the SYTO RNaselect (Figure 4G [\[link\]](#)) and RpoC-HaloTag-JF549 (Figure 4 – supplement 5C [\[link\]](#)) concentrations between the 1N and multi-N cells. However, to estimate the average 1N/multi-N SYTO RNaselect or RpoC-HaloTag-JF549 ratio during cell growth (Figure 4I [\[link\]](#) and Figure 4 – supplement 5E [\[link\]](#)) multiple samplings (with substitution) were performed for each biological replicate and cell area bin. This allowed us to use all the data while still comparing equal numbers of 1N and multi-N cells per biological replicate and cell area bin. Cell area bins with less than 10 cells per biological replicate were removed from the analysis. Also here, the sample size was set by the smallest population size (1N or multi-N population) and the number of iterations was equal to the difference between the populations multiplied by 10.

Single-molecule tracking analysis

Single-molecule tracking data was analyzed as previously described (Mäkelä and Sherratt, 2020 [\[link\]](#)). Candidate fluorescent spots were detected using band-pass filtering and an intensity threshold for each frame of the time-lapse sequence. These initial localizations were used as starting positions in phasor spot detection for high-precision localization (Martens et al., 2018 [\[link\]](#)). Individual molecules

were then tracked in each cell area by linking positions to a trajectory if they appeared in consecutive frames within a distance of 0.8 μm . Cell areas were detected from bright-field images using MicrobeTracker (Sliusarenko et al., 2011 [DOI](#)). In the case of multiple localizations within the tracking radius, these localizations were omitted from the analysis. Tracking allowed for a single frame disappearance of the molecule within a trajectory due to blinking or missed localization. The mobility of each molecule was determined by calculating an apparent diffusion coefficient, D_a , from the stepwise mean-squared displacement (MSD) of the trajectory using:

$$D_a = \frac{1}{4n\Delta t} \sum_{i=1}^n [x(i\Delta t) - x(i\Delta t + \Delta t)]^2 + [y(i\Delta t) - y(i\Delta t + \Delta t)]^2$$

where $x(t)$ and $y(t)$ indicate the coordinates of the molecule at time t , Δt is the frame rate of the camera, and n is the total number of the steps in the trajectory. Trajectories with less than 9 displacements were omitted due to the higher uncertainty in D_a .

The calculated D_a are expected to reflect different dynamic states of molecules. To determine the fraction of molecules in each state, $\log_{10}$ transformed D_a data (**Figure 2D** [DOI](#)) was fitted to a Gaussian mixture model (GMM) using the expectation-maximization algorithm (Bishop, Christopher, 2006). A mixture of 3 Gaussian distributions with free parameters for mean, SD, and weight of each state were fitted for different conditions (**Figure 2D** [DOI](#)). Additionally, for 1N cells, single-molecule tracking data were binned and fitted as a function of cell area (**Figure 2D - supplement 3** [DOI](#)). To determine the active RNAP or ribosome fraction of a single cell, the GMM was used to determine the state of each molecule from the measured D_a and the fraction of molecules in the slowest (“active”) state was calculated. Only cells with at least 50 trajectories were considered in the analysis for more accurate quantification. The total quantity of active molecules was estimated by multiplying the measured total fluorescence intensity with the measured active fraction as a function of the cell area.

Sample preparation for liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS)

CRISPRi *oriC* (SJ_XTL676) cells were collected 0, 120, 180, 240 and 300 min after addition of 0.2% arabinose, while CRISPRi *ftsZ* (SJ_XTL229) cells were collected after 0, 60, and 120 min of induction, and pelleted. Cell pellets were lysed in 1% SDS at 95°C for 10 min (with vigorous intermittent vortexing and in the presence of 5 mM β -mercaptoethanol as a reducing agent). Cell lysates were cleared by centrifugation at 15000 x g for 30 min at 4°C. The lysates were alkylated with 10 mM iodoacetamide for 15 min at room temperature, and then precipitated with three volumes of a solution containing 50% acetone and 50% ethanol. Precipitated proteins were re-solubilized in 2 M urea, 50 mM Tris-HCl, pH 8.0, and 150 mM NaCl, and then digested with TPCK-treated trypsin (50:1) overnight at 37°C. Trifluoroacetic acid was added to the digested peptides at a final concentration of 0.2%. Peptides were desalted with a Sep-Pak 50 mg C18 column (Waters). The C18 column was conditioned with five column volumes of 80% acetonitrile and 0.1% acetic acid, and then washed with five column volumes of 0.1% trifluoroacetic acid. After samples were loaded, the column was washed with five column volumes of 0.1% acetic acid followed by elution with four column volumes of 80% acetonitrile and 0.1% acetic acid. The elution was dried in a Concentrator at 45°C. Peptides (20 μg) resuspended in a 100-mM triethylammonium bicarbonate solution were labeled using 100 μg of Thermo TMT10plex™ in a reaction volume of 25 μl for 1 h. The labeling reaction was quenched with 8 μl of 5% hydroxylamine for 15 min. Labeled peptides were pooled, acidified to a pH of ~2 using drops of 10% trifluoroacetic acid, and desalted again with a Sep-Pak 50 mg C18 column as described above. TMT-labeled peptides were pre-fractionated using a Pierce™ High pH Reversed-Phase Peptide Fractionation Kit. Pre-fractionated peptides were dried using a concentrator and re-suspended in 0.1% formic acid.

LC-MS/MS data acquisition

Pre-fractionated TMT-labeled peptides were analyzed on a Fusion Lumos mass spectrometer (Thermo Fisher Scientific) equipped with a Thermo EASY-nLC 1200 LC system (Thermo Fisher Scientific). Peptides were separated by capillary reverse phase chromatography on a 25 cm column (75 μ m inner diameter, packed with 1.6 μ m C18 resin, AUR2-25075C18A, Ionopticks). Electrospray Ionization voltage was set to 1550 V. Peptides were introduced into the Fusion Lumos mass spectrometer using a 180 min stepped linear gradient at a flow rate of 300 nL/min. The steps of the gradient were as follows: 6–33% buffer B (0.1% (v/v) formic acid in 80% acetonitrile) for 145 min, 33–45% buffer B for 15 min, 40–95% buffer B for 5 min, and 90% buffer B for 5 min. Column temperature was maintained at 50°C throughout the procedure. Xcalibur software (Thermo Fisher Scientific) was used for the data acquisition and the instrument was operated in data-dependent mode. Advanced peak detection was disabled. Survey scans were acquired in the Orbitrap mass analyzer (centroid mode) over the range of 380 to 1400 m/z with a mass resolution of 120,000 (at m/z 200). For MS1 (the survey scan), the normalized AGC target (%) was set at 250 and the maximum injection time was set to 100 ms. Selected ions were fragmented by the Collision Induced Dissociation (CID) method with normalized collision energies of 34 and the tandem mass spectra were acquired in the ion trap mass analyzer with the scan rate set to “Rapid”. The isolation window was set to 0.7 m/z. For MS2 (the peptide fragmentation scan), the normalized AGC target (%) and the maximum injection time were set to “standard” and 35 ms, respectively. Repeated sequencing of peptides was kept to a minimum by dynamic exclusion of the sequenced peptides for 30 s. The maximum duty cycle length was set to 3 s. Relative changes in peptide concentration were determined at the level of the MS3 (reporter ion fragmentation scan) by isolating and fragmenting the five most dominant MS2 ion peaks.

Spectral searches

All raw files were searched using the Andromeda engine (Thomason, Costantino, and Court 2007) embedded in MaxQuant (v2.3.1.0) (Cox and Mann, 2008). A Reporter ion MS3 search was conducted using 10plex TMT isobaric labels. Variable modifications included oxidation (M) and protein N-terminal acetylation. Carbamidomethylation of cysteines was a fixed modification. The number of modifications per peptide was capped at five. Digestion was set to tryptic (proline-blocked). Database search was conducted using the UniProt proteome Ecoli_UP000000625_83333. The minimum peptide length was seven amino acids. The false discovery rate was determined using a reverse decoy proteome (Elias and Gygi, 2007).

Proteomics data analysis

Normalization and protein slope calculations were performed as described previously (Lanz et al., 2022). In brief, the relative signal difference between the TMT channels for each peptide was plotted against the normalized cell area for each of the bins of *E. coli* cells. For protein detection, we used a minimum of three unique peptide measurements per protein as a threshold. To derive the protein slope values shown in Figure 5, individual peptide measurements were consolidated into a protein level measurement using Python’s groupby.median. Peptides with the same amino acid sequence that were identified as different charge states or in different fractions were considered independent measurements. We summarized the size scaling behavior of individual proteins as a slope value derived from a regression. Each protein slope value was based on the behavior of all detected peptides.

For a given protein, we calculated the cell size-dependent slope as follows:

y_i = relative signal in the i^{th} TMT channel (median of all corresponding peptides in this channel)

x_i = normalized cell size in the i^{th} TMT channel (cell area for a given timepoint / mean cell area for the experiment)

The protein slope value was determined from a linear fit to the log-transformed data using the equation:

$$\text{Log}_2(y) = \text{Slope} * \text{log}_2(x)$$

Variables were log-transformed so that a slope of 1 corresponds to an increase in protein concentration that is proportional to the increase in cell volume, and a slope of -1 corresponds to 1/volume dilution. Pearson correlation coefficients and p values were calculated using SciPy's `pearsonr` module in Python (Virtanen et al., 2020 [DOI](#)). The results were reproducible across the two replicates (**Figure 5 – figure supplement 1** [DOI](#)).

Estimation of protein abundance using summed ion intensity

MS1-level peptide ion intensities from experiment #1 were used to estimate the relative protein abundance. For each protein, all peptide intensities were summed together using the “Intensity” column of MaxQuant's evidence.txt file. To adjust for the fact that larger proteins produce more tryptic peptides, the summed ion intensity for each protein was divided by its amino acid sequence length.

Calculation of distance from *oriC*

Gene coordinates were downloaded from EcoCyc database. The midpoint of each coding sequence was used to determine the circular distance (in base pairs) from *oriC* (base pair 3,925,859 of the total length of 4,641,652). A script was written to make three distance calculations. First, the direct distance (the midpoint of each gene minus 3,925,859 bp (*oriC*)) was determined. Next, the midpoint coordinate of each gene was subtracted by the total genome length (4,641,652 bp) and then subtracted again by the coordinate of *oriC* (3,925,859 bp). Finally, the midpoint coordinate of each gene was added to the total genome length (4,641,652 bp) and then subtracted by the coordinate of *oriC* (3,925,859 bp). The absolute values of these three calculations were then taken and the minimal values represent the circular distances between *oriC* and each gene.

Sample preparation for RNA-seq experiments

CRISPRi *oriC* strain (SJ_XTL676) was grown in M9glyCAAT at 37°C and cells were collected 60, 120, 200, and 240 min after addition of 0.2% L-arabinose. Images of cells were acquired at each timepoint to determine cell area, as described above. For each timepoint, the aliquot was spun down, the supernatant was removed, and the pellet was flash-frozen. *E. coli* pellets were resuspended in ice cold phosphate buffer saline solution (PBS) and mixed with *C. crescentus* cells in approximately a 1-to-1 ratio based on OD₆₀₀. *C. crescentus* cells were originally included with the intent of using them as a spike-in reference. However, the fraction of final reads obtained from *C. crescentus* transcripts was inconsistent with the initial mixing ratios. In addition, without knowing the exact DNA concentration in each 1N-rich cell sample, this spike-in became purposeless and thus was ignored for analysis. Cells (50 µL) in PBS were mixed with 250 µL TRI Reagent (Zymo Research) and lysed by bead beating on a Fastprep 24 (MPbio). Cell debris were pelleted (14k rpm, 2 min) and the supernatant was recovered. RNA was then extracted using the direct-zol RNA microprep kit (Zymo Research). Ribosomal RNA (rRNA) was then depleted using the NEBNext rRNA Depletion Kit for Bacteria (NEB, #E7850) and NEBNext Ultra II Directional RNA Library Prep Kit for Illumina (NEB, #E7760) was then used to prepare libraries for paired-end (2x150bp) Illumina sequencing (Novogene).

RNA-seq data analysis

A combined genome file of *E. coli* K-12 MG1655 and *C. crescentus* NA1000 with gene annotations was generated using a previously described approach (Swaffer et al., 2023). Read mapping statistics and genome browser tracks were generated using custom Python scripts. For quantification purposes, reads were aligned as 2x50mers in transcriptome space against an index generated from the combined gene annotation model using Bowtie (version 1.0.1; settings: -e 200 -a -X 1000) (Langmead et al., 2009). Alignments were then quantified using eXpress (version 1.5.1) (Roberts and Pachter, 2013) as TPM (Transcripts Per Million). TPM values were then recalculated after filtering for only *E. coli* genes. *E. coli* genes with TPM values below 1 were then removed from analysis.

The RNA slope (i.e., relative transcript concentration vs. cell area) was calculated for each gene as follows. TPM values were normalized to the mean of all values for that gene and then log2 transformed. The same normalization was applied to the cell area measurements for each condition. A linear model was fitted to the normalized log2 data and the slope of the linear regression was taken as the RNA slope.

Mathematical model and simulations

Phenomenological model: We developed a phenomenological model to illustrate cell growth dynamics. In our models, the number of mRNA (X) and the number of protein (Y) in the cell are described by the differential equations:

$$\frac{dX}{dt} = r_1 \alpha_{RNAP}(X, Y)Y - \delta X \quad (\text{Eq.1})$$

$$\frac{dY}{dt} = r_2 \alpha_{ribo}(X, Y)Y \quad (\text{Eq. 2})$$

where $r_1 = \frac{\text{bulk transcription rate}}{\text{total protein number}}$, $r_2 = \frac{\text{bulk translation rate}}{\text{total protein number}}$, $\alpha_{RNAP}(X, Y)$ is the fraction of active RNA polymerases, $\alpha_{ribo}(X, Y)$ is the fraction of active ribosomes, and, δ is the mRNA degradation rate. We assumed that the protein degradation is negligible because most bacterial proteins are stable (Balakrishnan et al., 2022; Lin and Amir, 2018). For the fraction of active RNAPs, we considered two different models (models A and B). In model A, the fraction of active RNAPs was modeled by $\alpha_{RNAP} = \frac{[Z]}{K_1 + [Z]}$. In model B, where we included different states of RNAP and promoter, the expression of α_{RNAP} is more complicated and is described in Appendix 2. For both models, the ribosomes were modeled by $\alpha_{ribo} = \frac{[X]}{K_2 + [X]}$. We used $[Z]$ and $[X]$ to represent the genome and mRNA concentrations, respectively. We assumed that the cell volume (V) and cell area (A) are proportional to the protein number Y , which is a good approximation (Basan et al., 2015; Kubitschek et al., 1984). While this assumption has not been verified in 1N cells, the concentration of ribosomes, which constitute most of the protein mass, were found to remain constant in these cells (Figure 2A). Namely, in our model, $V = cY$ and $A = c'Y$ with constants c, c' . Under balanced growth conditions, $[Z]$ was assumed to be constant. For the DNA-limited growth condition (as in 1N cells), we assumed that the genome does not duplicate and is kept at a single copy per cell ($(t) = 1$) while the cell volume continue to increase. The details for parameters used in this model are described in Table 1.

Numerical simulations: To simulate the model for balanced and DNA-limited growths, we first fixed a parameter set based on the literature (see Tables 1 and 2) and simulated Eq. 1 and Eq. 2 to reach the regime of balanced growth. Under balanced growth, we obtained the balanced growth vector (X^*, Y^*) where Y^* matches the initial protein number reported in the

literature (see **Table 2**). Then, using the balanced growth vector as the initial condition, we simulated the balanced growth and DNA-limited growth using different assumptions of genome content (t) (described in the previous paragraph). The simulation was performed using the MATLAB built-in function *ode23* with relative and absolute errors of the numerical values to be 10^{-4} and 10^{-5} , respectively. For each growth condition, we obtained growth trajectories $\mu(t)$, $Y(t)$, and from $Y(t)$, we calculated the cell volume $V(t)$ and area $A(t)$ as described above. For model A, the fraction of active RNAPs was calculated by $\alpha_{RNAP} = \frac{[Z]}{K_1 + [Z]}$, while for model B, it is calculated using a more complicated formula that take into account RNAP kinetics (see Appendix 2). For both models, the fraction of active ribosomes was calculated as $\alpha_{ribo} = \frac{[X]}{K_2 + [X]}$.

Parameter fitting: For either model A or B, we fitted our simulation results to all the experimental data including growth curves, the fraction of active RNAPs, and the fraction of active ribosomes for 1N and WT or multi-N cells (six datasets in total); see Appendix 1 for the initial parameter estimation and Appendix 3 for the optimization procedure details. The optimized parameter sets (see **Table 2**) were used for the numerical simulations shown in **Figure 4**. Note that the optimized parameter set remained within the realistic range for cellular physiology (see Appendix 3 - figure 1 for comparison between original and optimized parameters).

Data availability

Microscopy image analysis code is available on the Jacobs-Wagner lab github repository (https://github.com/JacobsWagnerLab/published/tree/master/Makela_2023). The TMT-MS and RNA-seq data analysis and processing code is available at https://github.com/mikechucklanz/Proteome_size_scaling_Ecoli and <https://github.com/georgimarinov/GeorgiScripts>.

For this study, the following TMT-MS and RNA-seq data were generated. Dataset 1 includes the calculated protein slopes from 1N-rich and multi-N cells, the normalized protein proportions, the summed ion intensities for each protein as well as the distance of its gene from the origin of replication (*oriC*) and the GO-term analysis. Dataset 2 includes the calculated mRNA slopes, in comparison with the average protein slopes from 1N-rich cells. All sequencing data associated with this study were submitted to the GEO repository on 2/10/2024 (GEO: GSEXXXX—will be updated when available). The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository with the dataset identifier PXD050093.

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Note

This reviewed preprint has been updated to add an equal contribution statement to the acknowledgements.

Supplementary figure legends

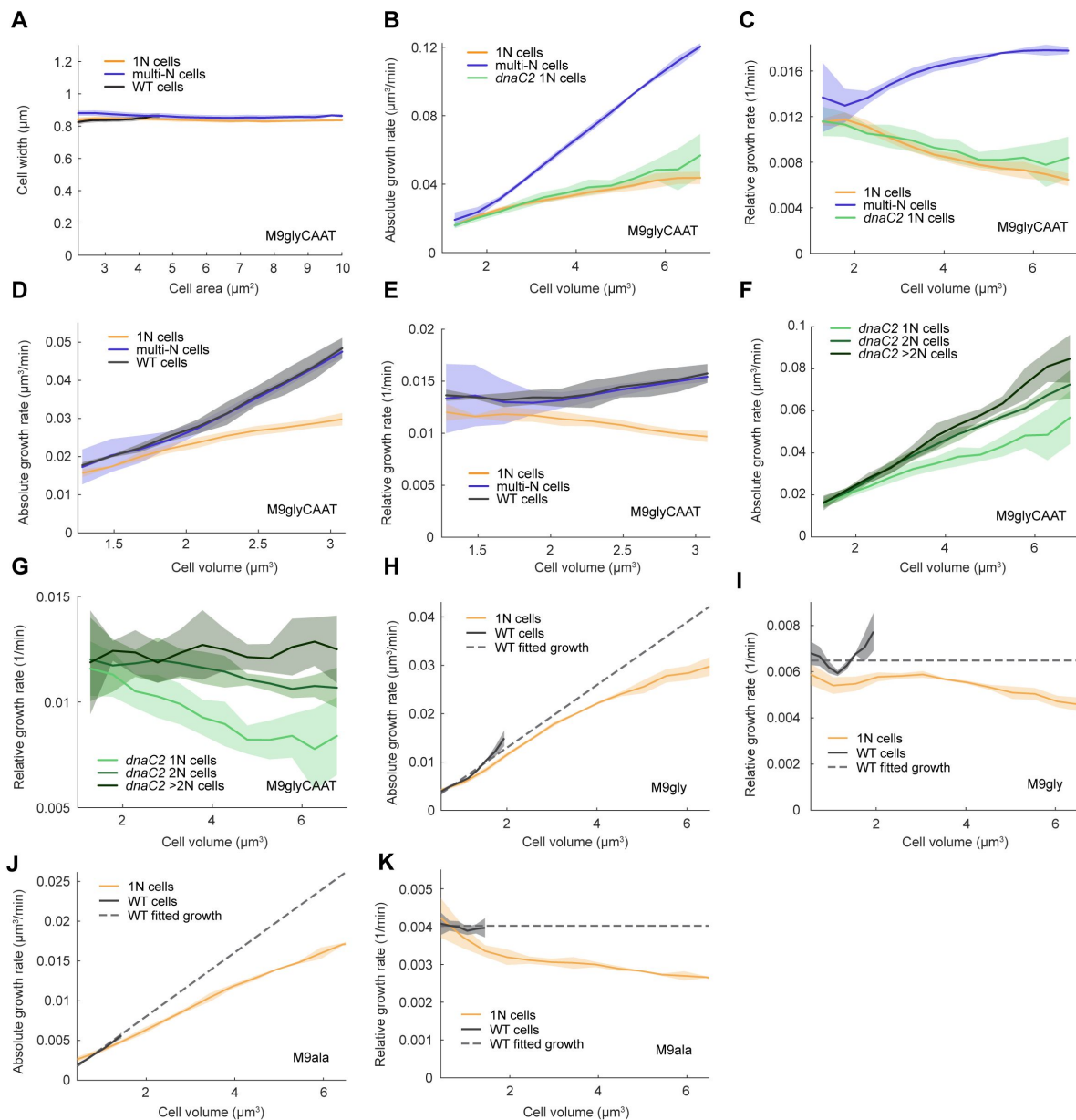


Figure 1 – figure supplement 1.

Relationships between growth rate and cell volume across several growth conditions and cell types of different ploidy.

(A) Cell width in WT (59072 datapoints from 7640 cells, CJW7339), 1N (41313 datapoints from 1585 cells, CJW7477), and multi-N (18489 datapoints from 915 cells, CJW7563) cells in M9glyCAAT as a function of cell area. Lines and shaded areas denote mean $\pm$ SD from 3 biological replicates. (B) Absolute and (C) relative growth rate based on cell volume in 1N (32735 datapoints from 1568 cells, CJW7457), multi-N (14006 datapoints from 916 cells, CJW7576), and *dnaC2* 1N (13933 datapoints from 1043 cells, CJW7374) cells as a function of cell volume in M9glyCAAT. Lines and shaded areas denote mean $\pm$ SD from three experiments. The growth rates are based on volume in this figure. (D) Absolute and (E) relative growth rate in M9glyCAAT in 1N (32735 datapoints from 1568 cells, CJW7457), multi-N (14006 datapoints from 916 cells, CJW7576) and WT (19495 datapoints from 7440 cells, CJW7339) cells as a function of cell area. (F) Absolute growth rate and (G) relative growth rate in 1N (13933 datapoints from 1043 cells), 2N (6265 datapoints from 295 cells) and >2N (2116 datapoints from 95 cells) *dnaC2* (CJW7374) cells as a function of cell area in M9glyCAAT. (H) Absolute and (I) relative growth rate in 1N (50352 from 973 cells, CJW7457) and WT (80269 datapoints from 12544 cells, CJW7339) cells in M9gly. (J) Absolute and (K) relative growth rate in 1N (71736 datapoints from 909 cells, CJW7457) and WT (63367 datapoints from 6880 cells, CJW7339) cells in M9ala. Lines and shaded areas denote mean $\pm$ SD from three biological replicates.

Figure 1 – figure supplement 2.

Relationships between growth rate and cell area in cell types of different ploidy.

(A) Absolute and (B) relative growth rate growth rate in M9glyCAAT in 1N (32735 datapoints from 1568 cells, CJW7457), multi-N (14006 datapoints from 916 cells, CJW7576) and WT (19495 datapoints from 7440 cells, CJW7339) cells as a function of cell area. Lines and shaded areas denote mean $\pm$ SD from three biological replicates.

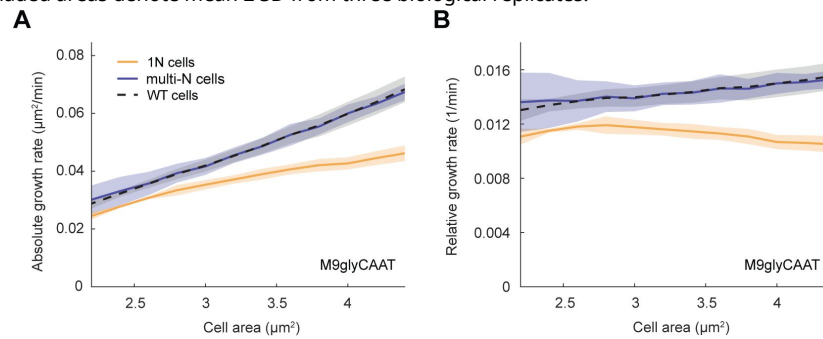


Figure 1 – figure supplement 3.

Characterization of ploidy in different cell types.

(A) Representative microscopy images of CRISPRi *oriC* cells expressing HU-CFP and ParB-mCherry with *parS* site at *ori1* (CJW7517) in M9glyCAAT 210 min after addition of 0.2% L-arabinose. Scale bar: 1 μm . (B) Representative microscopy images of WT (top) and *dnaC2* (bottom) cells expressing HU-mCherry growing under microscope observation in M9glyCAAT. DNA replication in *dnaC2* cells was blocked by growing cells at 37°C for 145 min. Arrows indicate *dnaC2* cells with one (1N) or two (2N) nucleoids. Scale bar: 1 μm . (C) Graph showing the percentage of *dnaC2* cells (CJW7374) with 1, 2 or >2 nucleoids after growth at 37°C. Shown are aggregated data from three biological replicates.

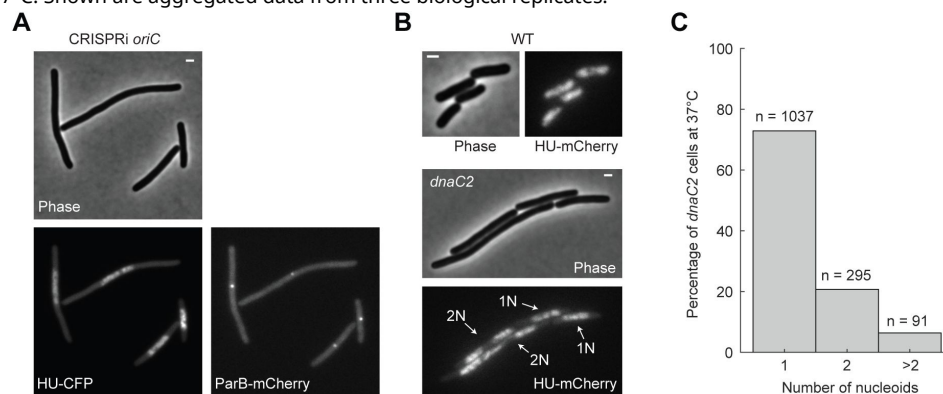


Figure 1 – figure supplement 4.

Determination of absolute growth rate of ppGpp⁰ (CJW7518) and $\Delta recA$ (CJW7522) cells grown in M9glyCAAT as a function of cell area.

Lines and shaded areas denote mean $\pm$ SD from two biological replicates. Also shown are the data for 1N (CJW7457) and multi-N (CJW7576) cells from [Figure 1C](#).

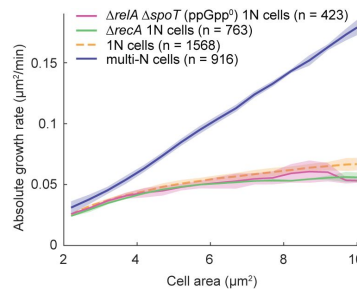


Figure 1 – figure supplement 5.

Validation of stable growth conditions under microscope observation.

(A) Plot showing the absolute growth rate of 1N (CJW7457) cells grown 90 min in a liquid M9glyCAAT culture in the presence of 0.2% L-arabinose and then transferred to an agarose pad and imaged for 2 h. The data for 1N (CJW7457) and multi-N (CJW7576) cells presented in [Figure 1C](#) are also shown for comparison; these cells were grown on an agarose pad for 6 h. Lines and shaded areas denote mean $\pm$ SD from two biological replicates. (B) Plot showing the relative growth rate of WT (CJW7339) cells in M9glyCAAT grown after placing them on an agarose pad. Lines and shaded areas denote mean $\pm$ SD from three biological replicates.

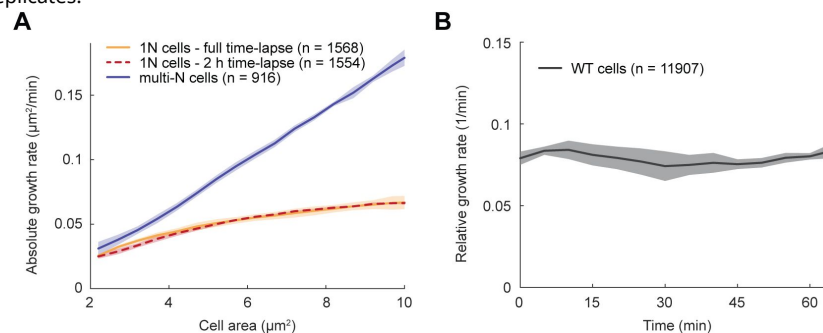


Figure 1 – figure supplement 6.

DNA-dependent growth in *Caulobacter crescentus*.

(A) Plot showing the absolute and (B) relative growth rate of 1N (DnaA depletion, strain CJW4823, 87 cells) and multi-N (FtsZ depletion, strain CJW3673, 181 cells) *C. crescentus* as a function of cell area. Lines and shaded areas denote mean $\pm$ SD from three biological replicates. (C) Example images of 1N and multi-N *C. crescentus* cells with ploidy distinguished by the number of ParB-eCFP foci. Scale bar: 2 μm .

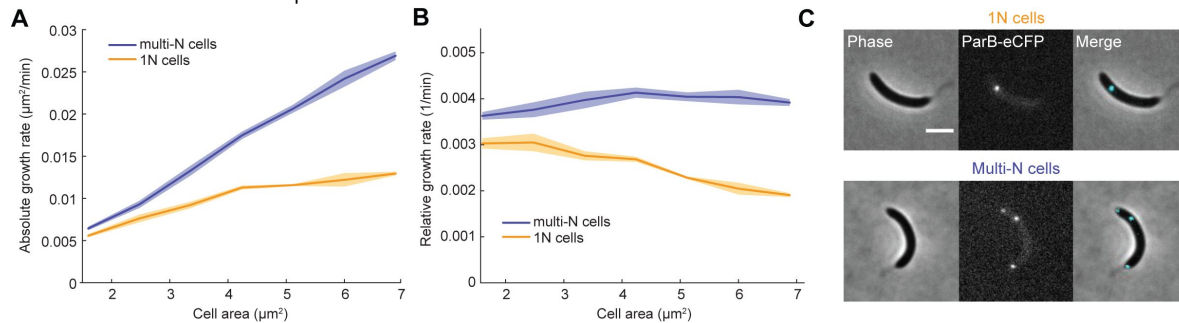


Figure 2 – figure supplement 1.

Determination of RpsB-msfGFP concentration in 1N cells.

Plot showing the RpsB-msfGFP fluorescence concentration in 1N (CJW7478) cells grown in (A) M9gly (from three experiments) or (B) M9ala (from two experiments) as a function of cell area. Lines and shaded areas denote mean $\pm$ SD between the biological replicates.

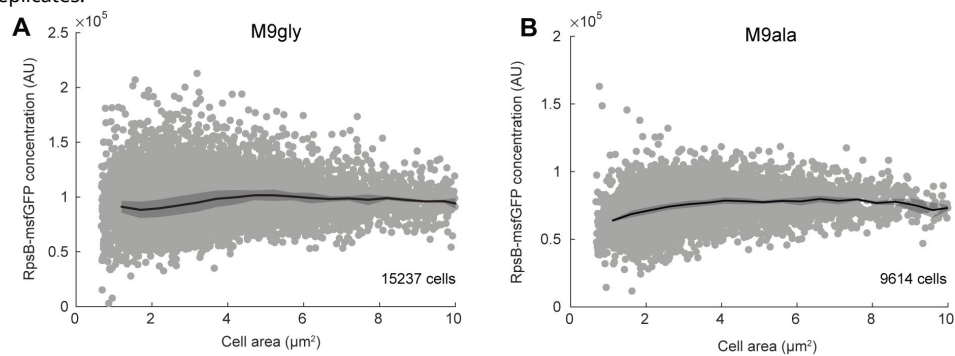


Figure 2 – figure supplement 2.

Diffusive characteristics of labeled ribosomes in rifampicin-treated WT cells.

(A) Plot showing the probability density of apparent diffusion coefficients (D_a) of JF549-labeled RpsB-HaloTag in WT cells (CJW7528) treated with 200 $\mu\text{g}/\text{mL}$ rifampicin for 30 min. Only tracks of length ≥ 10 are included. Also shown is D_a fitted by three-state Gaussian mixture model (GMM) (mean $\pm$ SEM). Data are from three biological replicates.

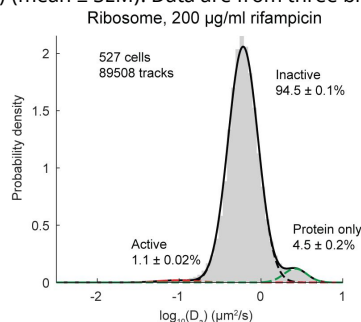
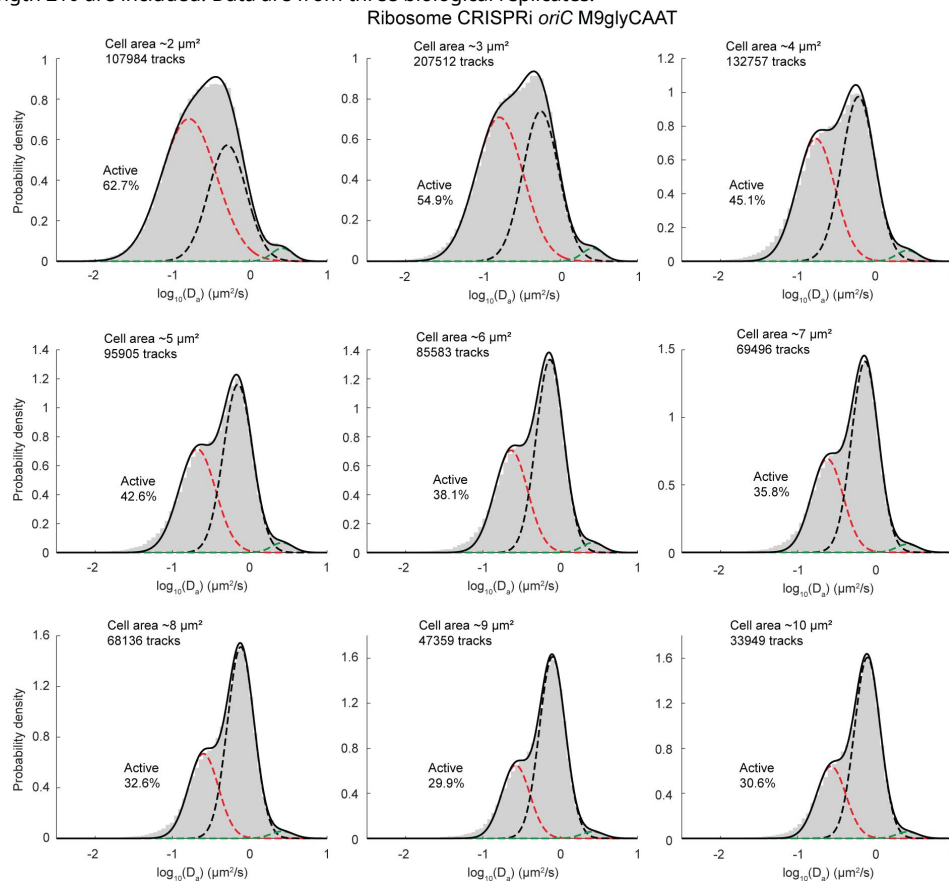


Figure 2 – figure supplement 3.

Diffusive characteristics of labeled ribosomes in 1N cells as a function of cell area.

Plots showing the probability density of apparent diffusion coefficients (D_a) of JF549-labeled RpsB-HaloTag in 1N cells (CJW7529) in M9glyCAAT at different cell areas. Also shown is D_a fitted by three-state Gaussian mixture model (GMM). Only tracks of length ≥ 10 are included. Data are from three biological replicates.



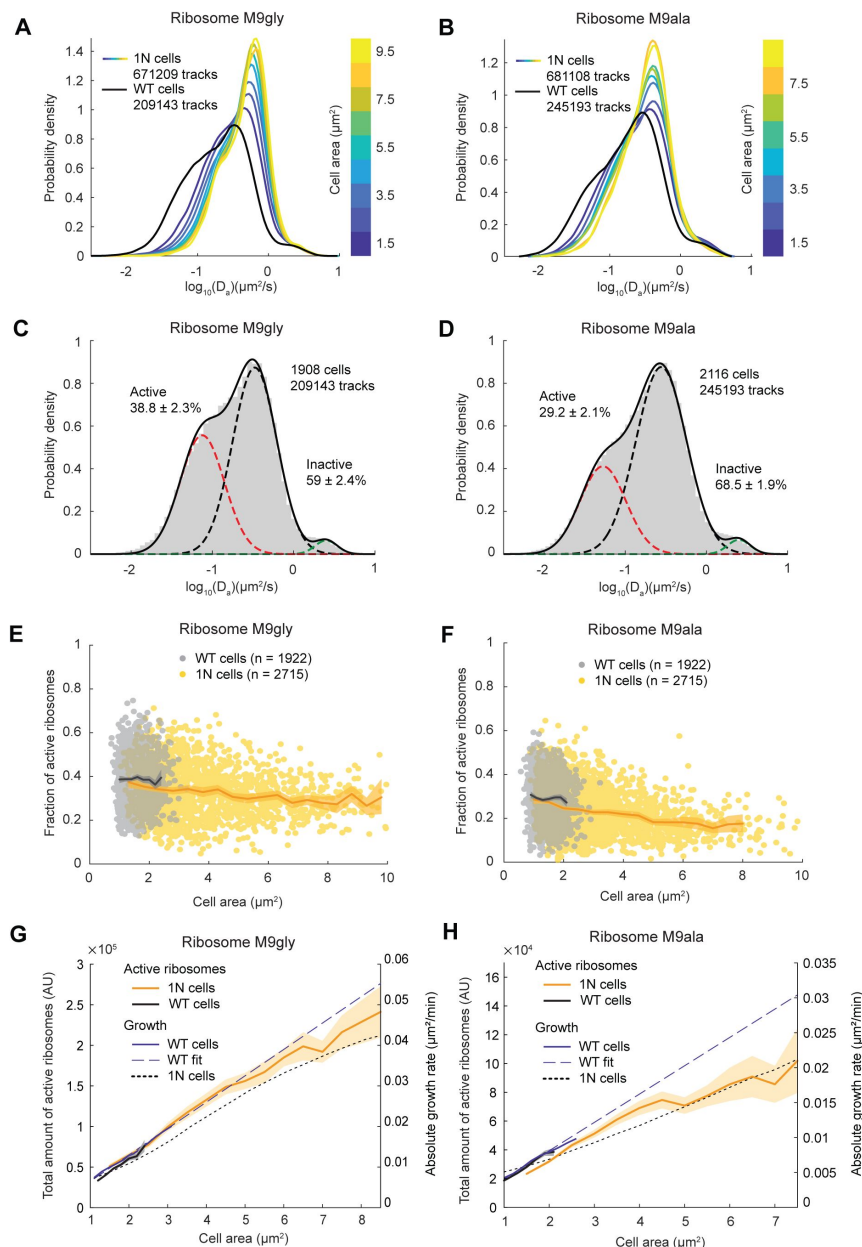


Figure 2 – figure supplement 4.

Characterization of ribosome diffusion and active fraction in poor media conditions.

(A) Plot showing the probability density of apparent diffusion coefficients (D_a) of JF549-labeled RpsB-HaloTag across cell areas for WT (CJW7528) and 1N (CJW7529) cells grown in M9gly. Only tracks of length ≥ 10 are included. 1N cells were binned according to cell area. (B) Same as (A) for cells grown in M9ala. (C) Plot showing probability density of D_a of JF549-labeled RpsB-HaloTag in WT cells (CJW7528) grown M9gly. Only tracks of length ≥ 10 are included. Also shown is D_a fitted by three-state Gaussian mixture model (GMM) (mean $\pm$ SEM). (D) Same as (C) but for cells grown in M9ala. (E) Plot showing the fraction of active ribosomes as a function of cell area for individual WT (CJW7528) and 1N (CJW7529) cells (dots) grown in M9gly. Only cells with ≥ 50 tracks are included. Shaded areas denote 95% confidence interval (CI) of the mean from bootstrapping. (F) same as (E) but for cells grown in M9ala. (G) Plot showing the total active ribosome amount (left y-axis) and the absolute growth rate (right y-axis) of 1N and multi-N cells (from [Figure 1C](#)) grown in M9gly as a function of cell area. The total amount of active ribosomes was calculated by multiplying the total amount of ribosomes by the fraction of active ribosomes. Also, shown is a linear fit to WT data ($f(x) = 2.99 \times 10^{-4}x$, $R^2 = 0.97$). Lines and shaded areas denote mean and 95% CI of the mean from bootstrapping. All data are from three biological replicates. (H) Same as (G) but for cells grown in M9Ala. Here the linear fit to the WT data is $f(x) = 1.90 \times 10^{-4}x$, $R^2 = 0.99$.

Figure 3 – figure supplement 1.

Determination of RpoC-YFP concentration in 1N cells.

(A) Plot showing the RpoC-YFP fluorescence concentration in 1N (CJW7477) cells grown in M9gly (three experiments) as a function of cell area. Lines and shaded areas denote mean $\pm$ SD between biological replicates. (B) Same as (A) but for 1N cells grown in M9ala (three experiments).

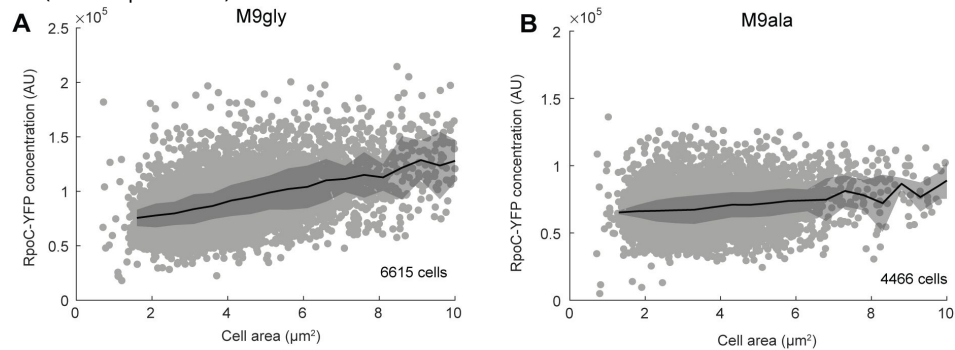
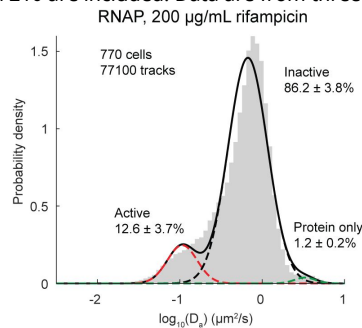


Figure 3 – figure supplement 2.

Diffusive characteristics of labeled RNAPs in rifampicin-treated WT cells.

Plot showing the probability density of apparent diffusion coefficients (D_a) of JF549-labeled RpoC-HaloTag in WT cells (CJW7519) treated with 200 $\mu\text{g}/\text{mL}$ rifampicin for 30 min. Also shown is D_a fitted by three-state Gaussian mixture model (GMM) (mean $\pm$ SEM). Only tracks of length ≥ 10 are included. Data are from three biological replicates.



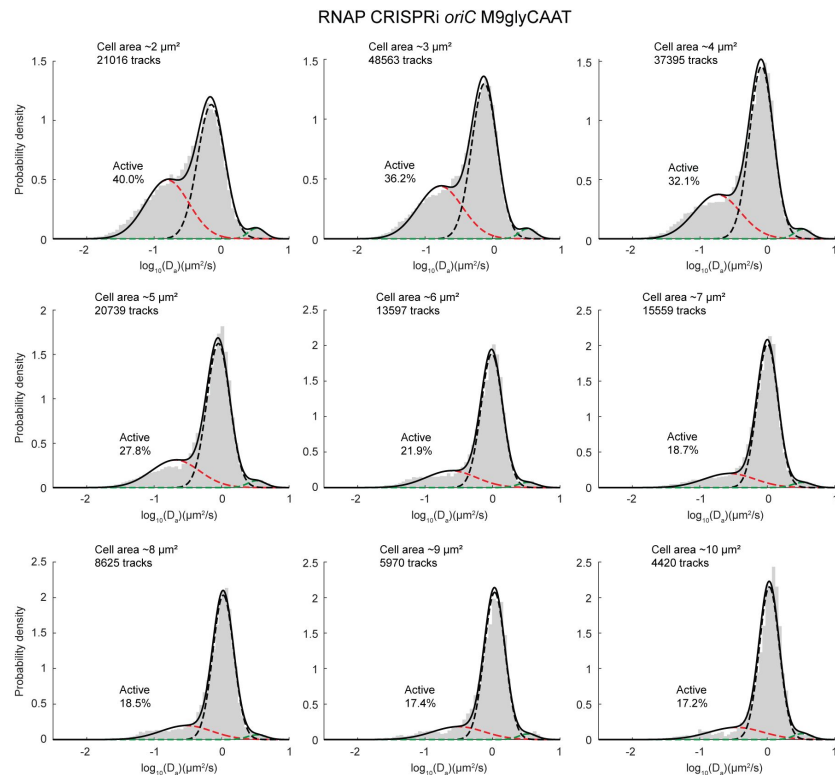


Figure 3 – figure supplement 3.

Diffusive characteristics of labeled RNAPs in 1N cells of different cell areas.

Plots showing the apparent diffusion coefficients (D_a) of JF549-labeled RpoC-HaloTag in 1N cells (CJW7520) of different cell areas. Cells were grown in M9glyCAAT. Also shown is D_a fitted by three-state Gaussian mixture model (GMM). Only tracks of length ≥ 10 are included. Data is from three biological replicates.

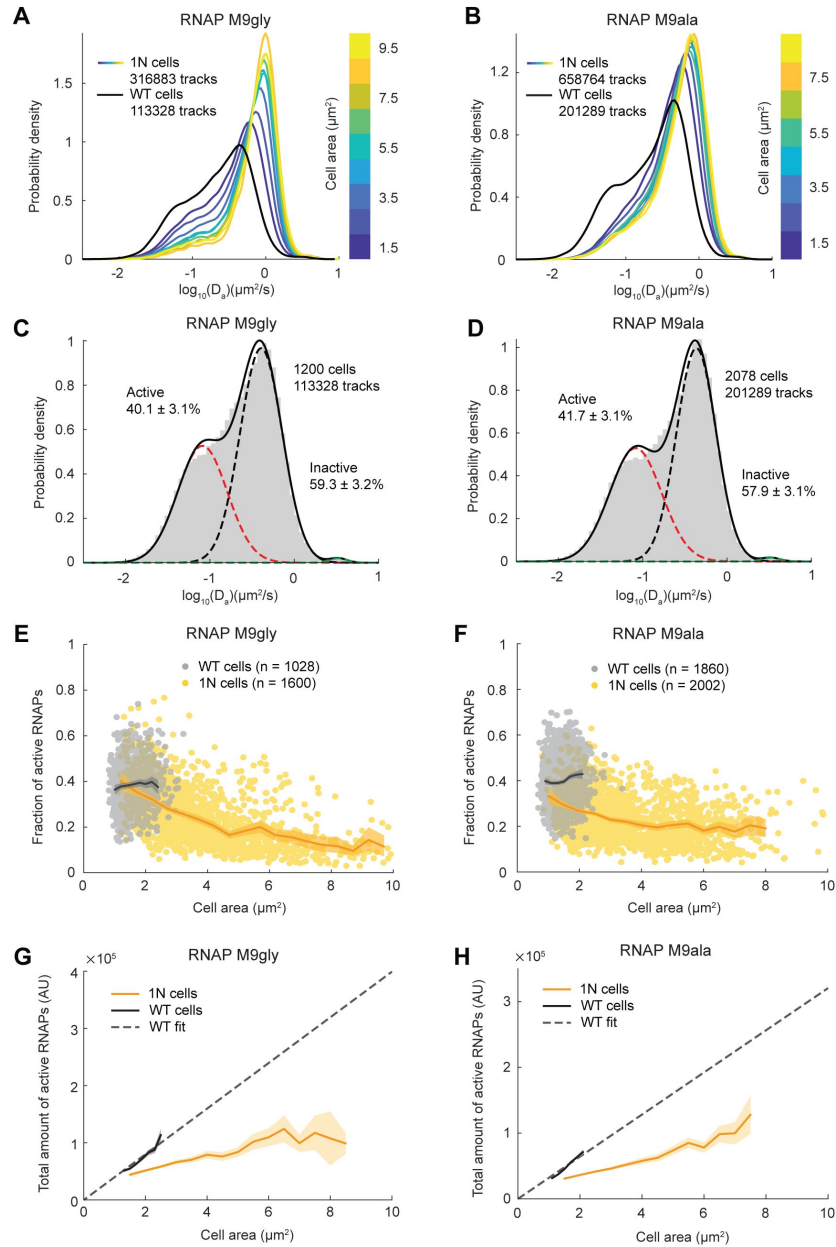


Figure 3 – figure supplement 4.

Characterization of RNAP diffusion and active fraction in poor media conditions.

(A) Plot showing the probability densities of apparent diffusion coefficients (D_a) of JF549-labeled RpoC-HaloTag in WT (CJW7519) and 1N (CJW7520) cells grown in M9gly. Only tracks of length ≥ 10 are included. 1N cells were binned according to cell area. (B) Same as (A) but for cells grown in M9ala. (C) Plot showing the probability density of D_a of JF549-labeled RpoC-HaloTag in WT cells (CJW7519) grown in M9gly. Also shown is D_a fitted by three-state Gaussian mixture model (GMM) (mean $\pm$ SEM). (D) Same as (C) but for cells grown in M9ala. (E) Plots showing the fraction of active RNAPs in individual WT (CJW7519) and 1N (CJW7520) cells (dots) grown in M9gly as a function of cell area. Only cells with ≥ 50 tracks are included. (F) Same as (E) but for cells grown in M9ala. (G) Plot showing the total amount of active RNAPs was calculated by multiplying the total amount of RNAPs by the fraction of active RNAPs in total and 1N cells grown in M9gly as a function of cell area. Also, shown is a linear fit to WT data ($f(x) = 3.99 \times 10^{-4} \times x$, $R^2 0.90$). Shaded areas denote 95% confidence interval (CI) of the mean from bootstrapping. All data are from three biological replicates. (H) Same as (G) but for cells grown in M9ala. Here, the linear fit for WT data is $f(x) = 3.21 \times 10^{-4} \times x$, $R^2 0.95$.

Figure 3 – figure supplement 5.

Determination of the relative Rsd concentration in 1N-rich and multi-N cells as a function of cell area.

Plots showing the relative protein concentration of Rsd in 1N-rich (SJ_XTL676) and multi-N (SJ_XTL229) cells as determined by TMT-MS. 1N-rich cells grown in M9glyCAAT were collected 0, 120, 180, 240 and 300 min, while multi-N cells were collected 0, 60, and 120 min after 0.2% L-arabinose induction of CRISPRi. Different shades of gray represent two independent experiments.

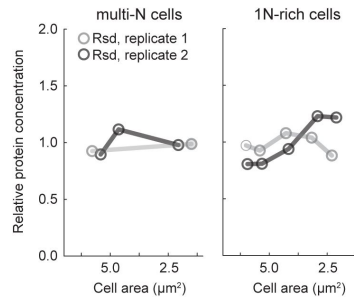


Figure 4 – figure supplement 1.

Comparison between experimental results from the M9glyCAAT condition and simulation results using model B.

(A-C) Plots comparing simulation results of model B (solid lines) with experimental data (dots) and averages (open squares) in the M9glyCAAT condition. The multi-N and 1N cells are indicated as blue and yellow, respectively: (A) The relation between the absolute growth rate ($\frac{dA}{dt}$) and cell area (A). (B) The relation between the active RNAP fraction and cell area. (C) The relation between the active ribosome fraction and cell area. (D) A two-dimensional diagram showing how the fractions of active RNAPs and ribosomes change with DNA concentration (colored from blue to yellow). Experimental data (with 2D error bars) from multi-N and 1N cells were combined and shown in the same plot.

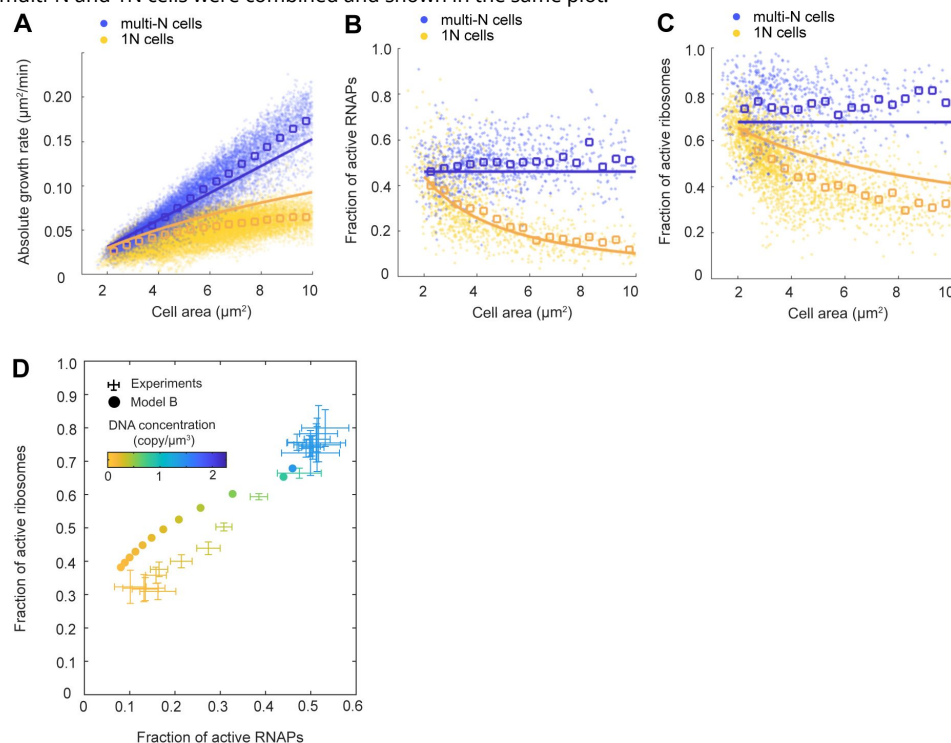


Figure 4 – figure supplement 2.

Comparison between experimental results from the M9gly and M9ala conditions and simulation results using model A.

(A-C) Plots comparing simulation results of model A (solid lines) with experimental data (dots) and averages (open squares) in the M9gly condition. The wild-type and 1N cells are indicated as black and yellow, respectively. (D-F) Plots comparing simulation results of model A (solid lines) with experimental data (dots) and averages (open squares) in the M9ala condition. The wild-type and 1N cells are indicated as black and yellow, respectively. (A,D) The relation between the absolute growth rate ($\frac{dA}{dt}$) and cell area (A). (B,E) The relation between the active RNAP fraction and cell area. (C,F) The relation between the active ribosome fraction and cell area.

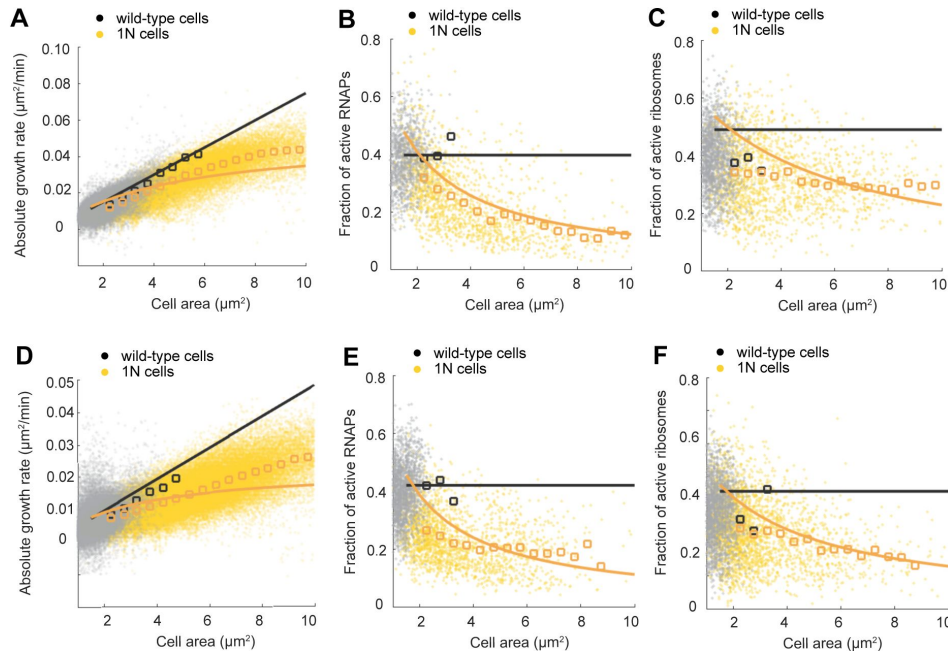
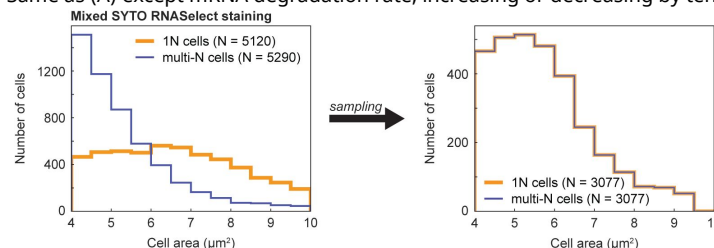


Figure 4 – figure supplement 3.

Model A-based simulations examining the effects of varying the rates in either mRNA synthesis or mRNA degradation on the relative growth rate of 1N cells as a function of cell area.

(A) Plot showing the decay of DNA concentration (black) and of the relative growth rate (blue) in 1N cells when the rate of bulk mRNA synthesis rate r_1 increase or decreases by ten-fold. Each quantity was normalized to its value at normal cell size (cell area = $2.5 \mu\text{m}^2$). (B) Same as (A) except mRNA degradation rate, increasing or decreasing by ten-fold.



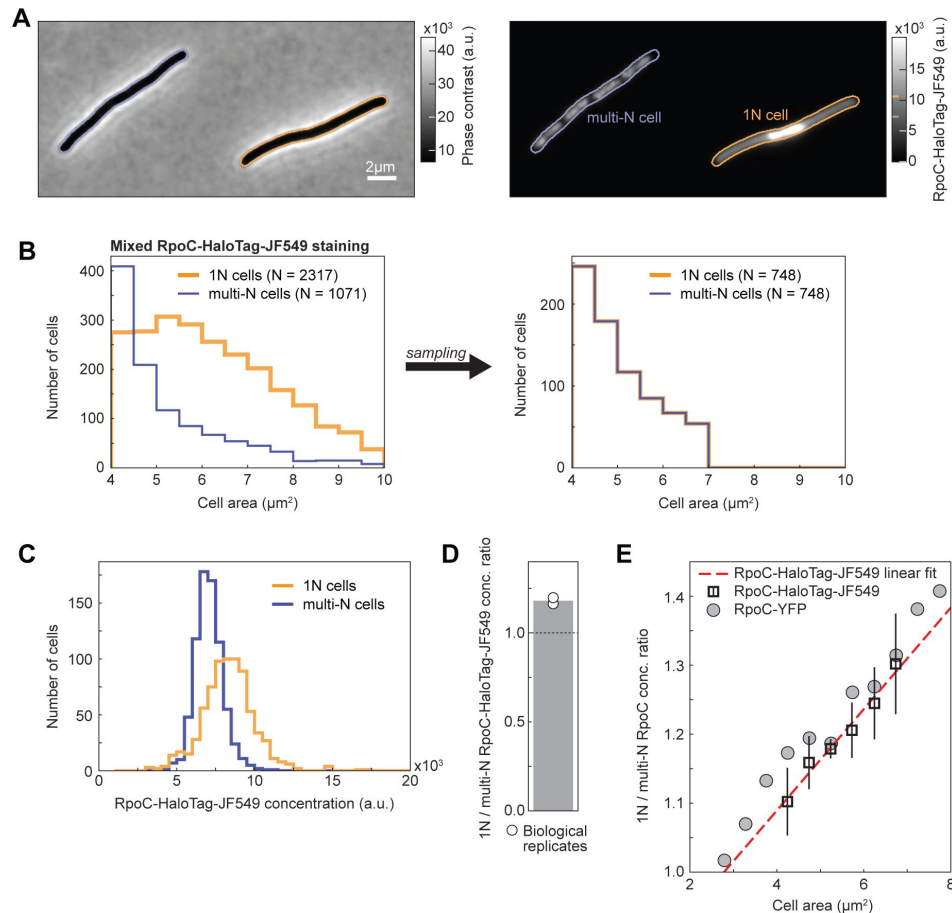


Figure 4 – figure supplement 4.

Cell sampling to match cell size distribution in mixed populations of 1N and multi-N cells.

Comparison of the cell area distributions of 1N and multi-N cells before and after random sampling. Single cells were sampled for each biological replicate and cell area bin such that the area distributions and numbers of 1N and multi-N cells overlapped. Bins with less than 50 cells from either population were removed from the analysis. Cell area distributions from mixed CJW7457 (CRISPRi *oriC*) and CJW7576 (CRISPRi *ftsZ*) populations stained with SYTO RNaselect (aggregated data from five biological replicates)

Figure 4 – figure supplement 5.

Comparison of RpoC-HaloTag-JF549 labelling between 1N and multi-N cells.

(A) Phase contrast (left) and RpoC-HaloTag-JF549 fluorescence (right) images of two representative cells from a mixed population of 1N (CRISPRi *oriC*, CJW7520) and multi-N (CRISPRi *ftsZ*, CJW7527) cells. The blue and orange ticks in the RpoC-HaloTag-JF549 color bar correspond to the average fluorescence (concentration proxy) of the multi-N and the 1N cell, respectively. (B) Same as Figure 4– figure supplement 4, except for CJW7520 (CRISPRi *oriC*) and CJW7527 (CRISPRi *ftsZ*) cells in which RpoC-HaloTag was stained with the JF549 dye (aggregated data from two biological replicates). (C) Distributions of RpoC-HaloTag-JF549 concentration for 1N and multi-N cells (748 cells for each population from two biological replicates). (D) Average 1N/multi-N RpoC-HaloTag-JF549 concentration ratio (gray bar) calculated from two biological replicates (white circles) after sampling the same number of cells per biological replicate and cell area bin for each population (see Materials and Methods). (E) Plot showing the concentration ratio of fluorescently labeled RpoC derivatives between 1N and multi-N cells versus the cell area. Squares show mean $\pm$ full range of RpoC-HaloTag-JF549 signal concentration ratios from two biological replicates, whereas grey circles indicate the mean signal concentration ratio of RpoC-YFP (from data shown in Figure 3A) for comparison. A linear regression (red dashed line) was fitted to the average RpoC-HaloTag-JF549 ratios. A cell area of $\sim 2.8 \mu\text{m}^2$ corresponds to a 1N/multi-N RpoC-HaloTag-JF549 concentration ratio equal to 1.

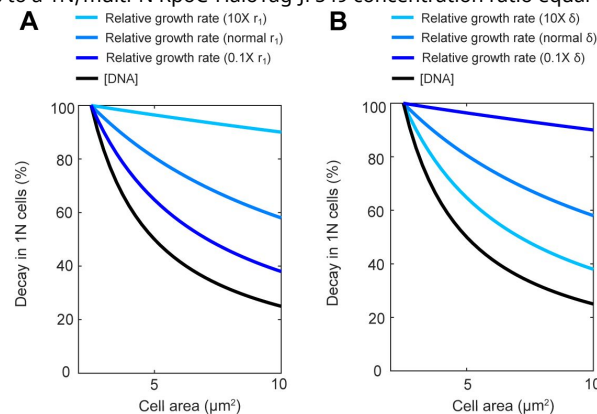
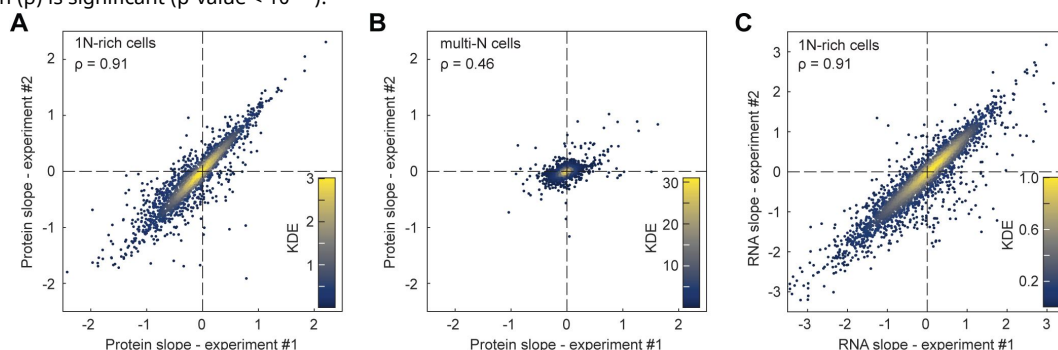


Figure 5 – figure supplement 1.

Comparison of protein and mRNA scaling between biological replicates.

(A) Correlation of protein slopes across the proteome (2360 proteins) between two biological replicates for 1N cells. The indicated Spearman correlation (ρ) is significant (p -value $< 10^{-10}$). (B) Same as (A) but for multi-N cells. (C) Correlation of RNA slopes across the genome (3446 mRNAs) between two biological replicates for 1N cells. The indicated Spearman correlation shown (ρ) is significant (p -value $< 10^{-10}$).



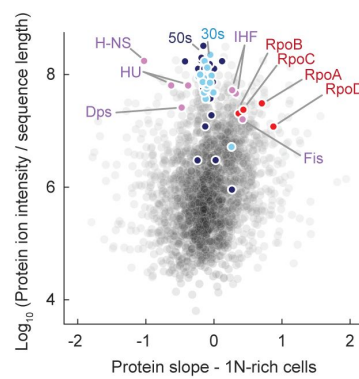


Figure 5 – figure supplement 2.

Protein slopes relative to protein ion intensity for 1N-rich cells.

The summed ion intensity of each protein was divided by the protein sequence length and the quotient was log-transformed. The locations of selected proteins are annotated.

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Reviewer #1 (Public Review):

Summary:

The manuscript by Mäkelä et al. presents compelling experimental evidence that the amount of chromosomal DNA can become limiting for the total rate of mRNA transcription and consequently protein production in the model bacterium *Escherichia coli*. Specifically, the authors demonstrate that upon inhibition of DNA replication the single-cell growth rate continuously decreases, in direct proportion to the concentration of active ribosomes, as measured indirectly by single-particle tracking. The decrease of ribosomal activity with filamentation, in turn, is likely caused by a decrease of the concentration of mRNAs, as suggested by an observed plateau of the total number of active RNA polymerases. These observations are compatible with the hypothesis that DNA limits the total rate of transcription and thus translation. The authors also demonstrate that the decrease of RNAP activity is independent of two candidate stress response pathways, the SOS stress response and the stringent response, as well as an anti-sigma factor previously implicated in variations of RNAP activity upon variations of nutrient sources.

Remarkably, the reduction of growth rate is observed soon after the inhibition of DNA replication, suggesting that the amount of DNA in wild-type cells is tuned to provide just as much substrate for RNA polymerase as needed to saturate most ribosomes with mRNAs. While previous studies of bacterial growth have most often focused on ribosomes and metabolic proteins, this study provides important evidence that chromosomal DNA has a previously underestimated important and potentially rate-limiting role for growth.

Strengths:

This article links the growth of single cells to the amount of DNA, the number of active ribosomes and to the number of RNA polymerases, combining quantitative experiments with theory. The correlations observed during depletion of DNA, notably in M9gluCAA medium, are compelling and point towards a limiting role of DNA for transcription and subsequently

for protein production soon after reduction of the amount of DNA in the cell. The article also contains a theoretical model of transcription-translation that contains a Michaelis-Menten type dependency of transcription on DNA availability and is fit to the data. While the model fits well with the continuous reduction of relative growth rate in rich medium (M9gluCAA), the behavior in minimal media without casamino acids is a bit less clear (see comments below).

At a technical level, single-cell growth experiments and single-particle tracking experiments are well described, suggesting that different diffusive states of molecules represent different states of RNAP/ribosome activities, which reflect the reduction of growth. However, I still have a few points about the interpretation of the data and the measured fractions of active ribosomes (see below).

Apart from correlations in DNA-deplete cells, the article also investigates the role of candidate stress response pathways for reduced transcription, demonstrating that neither the SOS nor the stringent response are responsible for the reduced rate of growth. Equally, the anti-sigma factor Rsd recently described for its role in controlling RNA polymerase activity in nutrient-poor growth media, seems also not involved according to mass-spec data. While other (unknown) pathways might still be involved in reducing the number of active RNA polymerases, the proposed hypothesis of the DNA substrate itself being limiting for the total rate of transcription is appealing.

Finally, the authors confirm the reduction of growth in the distant *Caulobacter crescentus*, which lacks overlapping rounds of replication and could thus have shown a different dependency on DNA concentration.

Weaknesses:

There are a range of points that should be clarified or addressed, either by additional experiments/analyses or by explanations or clear disclaimers.

First, the continuous reduction of growth rate upon arrest of DNA replication initiation observed in rich growth medium (M9gluCAA) is not equally observed in poor media. Instead, the relative growth rate is immediately/quickly reduced by about 10-20% and then maintained for long times, as if the arrest of replication initiation had an immediate effect but would then not lead to saturation of the DNA substrate. In particular, the long plateau of a constant relative growth rate in M9ala is difficult to reconcile with the model fit in Fig 4S2. Is it possible that DNA is not limiting in poor media (at least not for the cell sizes studied here) while replication arrest still elicits a reduction of growth rate in a different way? Might this have something to do with the naturally much higher oscillations of DNA concentration in minimal medium?

The authors argue that DNA becomes limiting in the range of physiological cell sizes, in particular for M9glCAA (Fig. 1BC). It would be helpful to know by how much (fold-change) the DNA concentration is reduced below wild-type (or multi-N) levels at $t=0$ in Fig 1B and how DNA concentration decays with time or cell area, to get a sense by how many-fold DNA is essentially 'overexpressed/overprovided' in wild-type cells.

Fig. 2: The distribution of diffusion coefficients of RpsB is fit to Gaussians on the log scale. Is this based on a model or on previous work or simply an empirical fit to the data? An exact analytical model for the distribution of diffusion constants can be found in the tool anaDDA by Vink, ..., Hohlbein *Biophys J* 2020. Alternatively, distributions of displacements are expressed analytically in other tools (e.g., in SpotOn).

The estimated fraction of active ribosomes in wild-type cells shows a very strong reduction with decreasing growth rate (down from 75% to 30%), twice as strong as measured in bulk experiments (Dai et al *Nat Microbiology* 2016; decrease from 90% to 60% for the same growth

rate range) and probably incompatible with measurements of growth rate, ribosome concentrations, and almost constant translation elongation rate in this regime of growth rates. Might the different diffusive fractions of RpsB not represent active/inactive ribosomes? See also the problem of quantification above. The authors should explain and compare their results to previous work.

To measure the reduction of mRNA transcripts in the cell, the authors rely on the fluorescent dye SYTO RNaselect. They argue that 70% of the dye signal represents mRNA. The argument is based on the previously observed reduction of the total signal by 70% upon treatment with rifampicin, an RNA polymerase inhibitor (Bakshi et al 2014). The idea here is presumably that mRNA should undergo rapid degradation upon rif treatment while rRNA or tRNA are stable. However, work from Hamouche et al. RNA (2021) 27:946 demonstrates that rifampicin treatment also leads to a rapid degradation of rRNA. Furthermore, the timescale of fluorescent-signal decay in the paper by Bakshi et al. (half life about 10min) is not compatible with the previously reported rapid decay of mRNA (2-4min) but rather compatible with the slower, still somewhat rapid, decay of rRNA reported by Hamouche et al.. A bulk method to measure total mRNA as in the cited Balakrishnan et al. (Science 2022) would thus be a preferred method to quantify mRNA. Alternatively, the authors could also test whether the mass contribution of total RNA remains constant, which would suggest that rRNA decay does not contribute to signal loss. However, since rRNA dominates total RNA, this measurement requires high accuracy. The authors might thus tone down their conclusions on mRNA concentration changes while still highlighting the compelling data on RNAP diffusion.

The proteomics experiments are a great addition to the single-cell studies, and the correlations between distance from ori and protein abundance is compelling. However, I was missing a different test, the authors might have already done but not put in the manuscript: If DNA is indeed limiting the initiation of transcription, genes that are already highly transcribed in non-perturbed conditions might saturate fastest upon replication inhibition, while genes rarely transcribed should have no problem to accommodate additional RNA polymerases. One might thus want to test, whether the (unperturbed) transcription initiation rate is a predictor of changes in protein composition. This is just a suggestion the authors may also ignore, but since it is an easy analysis, I chose to mention it here.

Related to the proteomics, in l. 380 the authors write that the reduced expression close to the ori might reflect a gene-dosage compensatory mechanism. I don't understand this argument. Can the authors add a sentence to explain their hypothesis?

In Fig. 1E the authors show evidence that growth rate increases with cell length/area. While this is not a main point of the paper it might be cited by others in the future. There are two possible artifacts that could influence this experiment: a) segmentation: an overestimation of the physical length of the cell based on phase-contrast images (e.g., 200 nm would cause a 10% error in the relative rate of 2 μ m cells, but not of longer cells). b) time-dependent changes of growth rate, e.g., due to change from liquid to solid or other perturbations. To test for the latter, one could measure growth rate as a function of time, restricting the analysis to short or long cells, or measuring growth rate for short/long cells at selected time points. For the former, I recommend comparison of phase-contrast segmentation with FM4-64-stained cell boundaries.

<https://doi.org/10.7554/eLife.97465.1.sa2>

Reviewer #2 (Public Review):

In this work, the authors uncovered the effects of DNA dilution on *E. coli*, including a decrease in growth rate and a significant change in proteome composition. The authors demonstrated that the decline in growth rate is due to the reduction of active ribosomes and

active RNA polymerases because of the limited DNA copy numbers. They further showed that the change in the DNA-to-volume ratio leads to concentration changes in almost 60% of proteins, and these changes mainly stem from the change in the mRNA levels.

<https://doi.org/10.7554/eLife.97465.1.sa1>

Reviewer #3 (Public Review):

Summary:

Mäkelä et al. here investigate genome concentration as a limiting factor on growth. Previous work has identified key roles for transcription (RNA polymerase) and translation (ribosomes) as limiting factors on growth, which enable an exponential increase in cell mass. While a potential limiting role of genome concentration under certain conditions has been explored theoretically, Mäkelä et al. here present direct evidence that when replication is inhibited, genome concentration emerges as a limiting factor.

Strengths:

A major strength of this paper is the diligent and compelling combination of experiment and modeling used to address this core question. The use of origin- and *ftsZ*-targeted CRISPRi is a very nice approach that enables dissection of the specific effects of limiting genome dosage in the context of a growing cytoplasm. While it might be expected that genome concentration eventually becomes a limiting factor, what is surprising and novel here is that this happens very rapidly, with growth transitioning even for cells within the normal length distribution for *E. coli*. Fundamentally, it demonstrates the fine balance of bacterial physiology, where the concentration of the genome itself (at least under rapid growth conditions) is no higher than it needs to be.

Weaknesses:

One limitation of the study is that genome concentration is largely treated as a single commodity. While this facilitates their modeling approach, one would expect that the growth phenotypes observed arise due to copy number limitation in a relatively small number of rate-limiting genes. The authors do report shifts in the composition of both the proteome and the transcriptome in response to replication inhibition, but while they report a positional effect of distance from the replication origin (reflecting loss of high-copy, origin-proximal genes), other factors shaping compositional shifts and their functional effects on growth are not extensively explored. This is particularly true for ribosomal RNA itself, which the authors assume to grow proportionately with protein. More generally, understanding which genes exert the greatest copy number-dependent influence on growth may aid both efforts to enhance (biotechnology) and inhibit (infection) bacterial growth.

Overall, this study provides a fundamental contribution to bacterial physiology by illuminating the relationship between DNA, mRNA, and protein in determining growth rate. While coarse-grained, the work invites exciting questions about how the composition of major cellular components is fine-tuned to a cell's needs and which specific gene products mediate this connection. This work has implications not only for biotechnology, as the authors discuss, but potentially also for our understanding of how DNA-targeted antibiotics limit bacterial growth.

<https://doi.org/10.7554/eLife.97465.1.sa0>

Author response:

eLife assessment

This study represents a fundamental contribution to our understanding of how gene expression levels are controlled in bacteria. Through a series of compelling and careful experiments, relying on a mutant that blocks DNA replication but permits growth, and using various methods, the authors reveal how genome concentration rapidly becomes limiting for growth when replication is inhibited. This work contributes to our understanding of the contributions and limiting roles of DNA, mRNA, and ribosomes for growth in bacteria, and will be of considerable interest within both systems biology and microbial physiology.

Thank you!

Public Reviews:**Reviewer #1 (Public Review):***Summary:*

*The manuscript by Mäkelä et al. presents compelling experimental evidence that the amount of chromosomal DNA can become limiting for the total rate of mRNA transcription and consequently protein production in the model bacterium *Escherichia coli*. Specifically, the authors demonstrate that upon inhibition of DNA replication the single-cell growth rate continuously decreases, in direct proportion to the concentration of active ribosomes, as measured indirectly by single-particle tracking. The decrease of ribosomal activity with filamentation, in turn, is likely caused by a decrease of the concentration of mRNAs, as suggested by an observed plateau of the total number of active RNA polymerases. These observations are compatible with the hypothesis that DNA limits the total rate of transcription and thus translation. The authors also demonstrate that the decrease of RNAP activity is independent of two candidate stress response pathways, the SOS stress response and the stringent response, as well as an anti-sigma factor previously implicated in variations of RNAP activity upon variations of nutrient sources.*

Remarkably, the reduction of growth rate is observed soon after the inhibition of DNA replication, suggesting that the amount of DNA in wild-type cells is tuned to provide just as much substrate for RNA polymerase as needed to saturate most ribosomes with mRNAs. While previous studies of bacterial growth have most often focused on ribosomes and metabolic proteins, this study provides important evidence that chromosomal DNA has a previously underestimated important and potentially rate-limiting role for growth.

Thank you for the excellent summary of our work.

Strengths:

This article links the growth of single cells to the amount of DNA, the number of active ribosomes and to the number of RNA polymerases, combining quantitative experiments with theory. The correlations observed during depletion of DNA, notably in M9gluCAA medium, are compelling and point towards a limiting role of DNA for transcription and subsequently for protein production soon after reduction of the amount of DNA in the cell. The article also contains a theoretical model of transcription-translation that contains a Michaelis-Menten type dependency of transcription on DNA availability and is fit to the data. While the model fits well with the continuous reduction of relative growth rate in rich medium (M9gluCAA), the behavior in minimal media without casamino acids is a bit less clear (see comments below).

At a technical level, single-cell growth experiments and single-particle tracking experiments are well described, suggesting that different diffusive states of molecules represent different states of RNAP/ribosome activities, which reflect the reduction of growth. However, I still have a few points about the interpretation of the data and the measured fractions of active ribosomes (see below).

Apart from correlations in DNA-deplete cells, the article also investigates the role of candidate stress response pathways for reduced transcription, demonstrating that neither the SOS nor the stringent response are responsible for the reduced rate of growth. Equally, the anti-sigma factor Rsd recently described for its role in controlling RNA polymerase activity in nutrient-poor growth media, seems also not involved according to mass-spec data. While other (unknown) pathways might still be involved in reducing the number of active RNA polymerases, the proposed hypothesis of the DNA substrate itself being limiting for the total rate of transcription is appealing.

*Finally, the authors confirm the reduction of growth in the distant *Caulobacter crescentus*, which lacks overlapping rounds of replication and could thus have shown a different dependency on DNA concentration.*

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We note that the total RNAP activity (abundance x active fraction) was also significantly reduced in poor media (Figure 3 -- supplement 4G and H) similarly to rich medium (Figure 3H). This is consistent with DNA being limiting. The main difference between rich and poor medium conditions is that the total ribosome activity in poor media (Figure 2 -- supplement 4G and H) was less affected in comparison to rich media (Figure 2H). Our interpretation of these results is that while DNA is limiting in all medium conditions (as shown by the RNAP data), changes in ribosome activity or mRNA degradation can compensate for the reduction in transcription in poor media and hence maintain better scaling of growth rates under DNA limitation. We understand how our current presentation made it confusing. We will reorganize the text and figures to better explain our results and interpretations.

The authors argue that DNA becomes limiting in the range of physiological cell sizes, in particular for M9glCAA (Fig. 1BC). It would be helpful to know by how much (fold-change) the DNA concentration is reduced below wild-type (or multi-N) levels at t=0 in Fig 1B and how DNA concentration decays with time or cell area, to get a sense by how many-fold DNA is essentially 'overexpressed/overprovided' in wild-type cells.

We will provide an estimate.

Fig. 2: The distribution of diffusion coefficients of RpsB is fit to Gaussians on the log scale. Is this based on a model or on previous work or simply an empirical fit to the data? An exact analytical model for the distribution of diffusion constants can be found in the tool anaDDA by Vink, ..., Hohlbein Biophys J 2020. Alternatively, distributions of displacements are expressed analytically in other tools (e.g., in SpotOn).

We use an empirical fit of Gaussian mixture model (GMM) of three states to the data and extract the fractions of molecules in each state. This avoids making too many assumptions on the underlying processes, e.g. a Markovian system with Brownian diffusion. The model in anaDDA (Vink et al.) is currently limited to two-transitioning states with a maximal step number of 8 steps per track for a computationally efficient solution (longer tracks are truncated). Using a short subset of the trajectories is less accurate than using the entire trajectory and because of this, we consider full tracks with at least 9 displacements. Meanwhile, Spot-On supports a three-state model but it is still based on a semi-analytical model with a pre-calculated library of parameters created by fitting of simulated data. Neither of these models considers the effect of cell confinement, which plays a major role on single-molecule diffusion in small-sized cells such as bacteria. For these reasons, we opted to use an empirical fit to the data. We note that the fractions of active ribosomes in WT cells grown in different media, which we extracted from these diffusion measurements, are consistent with estimates obtained by others using similar or different approaches (Forchhammer and Lindhal 1971; Mohapatra and Weisshaar, 2018; Sanamrad et al., 2014).

The estimated fraction of active ribosomes in wild-type cells shows a very strong reduction with decreasing growth rate (down from 75% to 30%), twice as strong as measured in bulk experiments (Dai et al Nat Microbiology 2016; decrease from 90% to 60% for the same growth rate range) and probably incompatible with measurements of growth rate, ribosome concentrations, and almost constant translation elongation rate in this regime of growth rates. Might the different diffusive fractions of RpsB not represent active/inactive ribosomes? See also the problem of quantification above. The authors should explain and compare their results to previous work.

We agree that our measured range is somewhat larger than the estimated range from Dai et al, 2016. However, they use different media, strains, and growth conditions. We also note that Dai et al did not make actual measurements of the active ribosome fraction. Instead, they calculate the “active ribosome equivalent” based on a model that includes growth rate, protein synthesis rate, RNA/protein abundance, and the total number of amino acids in all proteins in the cell. Importantly, our measurements show the same overall trend as Dai et al, 2016. Furthermore, our results are in quantitative agreements with previous experimental measurements that use ribosome profiling (Forchhammer and Lindhal 1971) or single-ribosome tracking (Mohapatra and Weisshaar, 2018; Sanamrad et al., 2014), which, we believe, validates our approach. We will clarify this point in the revised manuscript.

To measure the reduction of mRNA transcripts in the cell, the authors rely on the fluorescent dye SYTO RNaselect. They argue that 70% of the dye signal represents mRNA. The argument is based on the previously observed reduction of the total signal by 70% upon treatment with rifampicin, an RNA polymerase inhibitor (Bakshi et al 2014). The idea here is presumably that mRNA should undergo rapid degradation upon rif treatment while rRNA or tRNA are stable. However, work from Hamouche et al. RNA (2021) 27:946 demonstrates that rifampicin treatment also leads to a rapid degradation of rRNA. Furthermore, the timescale of fluorescent-signal decay in the paper by Bakshi et al. (half life about 10min) is not compatible with the previously reported rapid decay of mRNA (24min) but rather compatible with the slower, still somewhat rapid, decay of rRNA reported by Hamouche et al.. A bulk method to measure total mRNA as in the cited

Balakrishnan et al. (Science 2022) would thus be a preferred method to quantify mRNA. Alternatively, the authors could also test whether the mass contribution of total RNA remains constant, which would suggest that rRNA decay does not contribute to signal loss. However, since rRNA dominates total RNA, this measurement requires high accuracy. The authors might thus tone down their conclusions on mRNA concentration changes while still highlighting the compelling data on RNAP diffusion.

Thank you for bringing the Hamouche et al 2022 paper to our attention. We will address this point in the revised manuscript.

The proteomics experiments are a great addition to the single-cell studies, and the correlations between distance from ori and protein abundance is compelling. However, I was missing a different test, the authors might have already done but not put in the manuscript: If DNA is indeed limiting the initiation of transcription, genes that are already highly transcribed in non-perturbed conditions might saturate fastest upon replication inhibition, while genes rarely transcribed should have no problem to accommodate additional RNA polymerases. One might thus want to test, whether the (unperturbed) transcription initiation rate is a predictor of changes in protein composition. This is just a suggestion the authors may also ignore, but since it is an easy analysis, I chose to mention it here.

Thank you for the suggestion. We will provide the suggested analysis in the revised manuscript.

Related to the proteomics, in l. 380 the authors write that the reduced expression close to the ori might reflect a gene-dosage compensatory mechanism. I don't understand this argument. Can the authors add a sentence to explain their hypothesis?

We apologize for the confusion. This will be addressed in the revised manuscript.

In Fig. 1E the authors show evidence that growth rate increases with cell length/area. While this is not a main point of the paper it might be cited by others in the future. There are two possible artifacts that could influence this experiment: a) segmentation: an overestimation of the physical length of the cell based on phase-contrast images (e.g., 200 nm would cause a 10% error in the relative rate of 2 μ m cells, but not of longer cells). b) time-dependent changes of growth rate, e.g., due to change from liquid to solid or other perturbations. To test for the latter, one could measure growth rate as a function of time, restricting the analysis to short or long cells, or measuring growth rate for short/long cells at selected time points. For the former, I recommend comparison of phasecontrast segmentation with FM4-64-stained cell boundaries.

As the reviewer notes, the small increase in relative growth was just a minor observation that does not affect our story whether it is biologically meaningful or the result of a technical artefact. But we agree with the reviewer that others might cite it in future works and thus should be interpreted with caution.

An artefact associated with time-dependent changes (e.g. changing from liquid cultures to more solid agarose pads) is unlikely for two reasons. 1. We show that varying the time that cells spend on agarose pads relative to liquid cultures does not affect the cell size-dependent growth rate results (Figure 1 – supplement 5B). 2. We show that the growth rate is stable from the beginning of the time-lapse with no transient effects upon cell placement on agarose pads for imaging (Figure 1 – supplement 5B). These results were described in the Methods section where they could easily be missed. We will revise the text to discuss these controls more prominently in the Results section.

As for cell segmentation, we have run simulations and agree with the reviewer that a small overestimation of cell area (which is possible with any cell segmentation methods including ours) could lead to a small increase in relative growth with increasing cell areas. Since the finding is not important to our story, we will simply alert the readers to the possibility that the observation may be due to a small cell segmentation bias.

Reviewer #2 (Public Review):

In this work, the authors uncovered the effects of DNA dilution on E. coli, including a decrease in growth rate and a significant change in proteome composition. The authors demonstrated that the decline in growth rate is due to the reduction of active ribosomes and active RNA polymerases because of the limited DNA copy numbers. They further showed that the change in the DNA-to-volume ratio leads to concentration changes in almost 60% of proteins, and these changes mainly stem from the change in the mRNA levels.

Thank you for the support and accurate summary!

Reviewer #3 (Public Review):

Summary:

Mäkelä et al. here investigate genome concentration as a limiting factor on growth. Previous work has identified key roles for transcription (RNA polymerase) and translation (ribosomes) as limiting factors on growth, which enable an exponential increase in cell mass. While a potential limiting role of genome concentration under certain conditions has been explored theoretically, Mäkelä et al. here present direct evidence that when replication is inhibited, genome concentration emerges as a limiting factor.

Strengths:

A major strength of this paper is the diligent and compelling combination of experiment and modeling used to address this core question. The use of origin- and ftsZ-targeted CRISPRi is a very nice approach that enables dissection of the specific effects of limiting genome dosage in the context of a growing cytoplasm. While it might be expected that genome concentration eventually becomes a limiting factor, what is surprising and novel here is that this happens very rapidly, with growth transitioning even for cells within the normal length distribution for E. coli. Fundamentally, it demonstrates the fine balance of bacterial physiology, where the concentration of the genome itself (at least under rapid growth conditions) is no higher than it needs to be.

Weaknesses:

One limitation of the study is that genome concentration is largely treated as a single commodity. While this facilitates their modeling approach, one would expect that the growth phenotypes observed arise due to copy number limitation in a relatively small number of rate-limiting genes. The authors do report shifts in the composition of both the proteome and the transcriptome in response to replication inhibition, but while they report a positional effect of distance from the replication origin (reflecting loss of high-copy, origin-proximal genes), other factors shaping compositional shifts and their functional effects on growth are not extensively explored. This is particularly true for ribosomal RNA itself, which the authors assume to grow proportionately with protein. More generally, understanding which genes exert the greatest copy number-dependent influence on growth may aid both efforts to enhance (biotechnology) and inhibit (infection) bacterial growth.

We agree but feel that identifying the specific limiting genes is beyond the scope of the study. However, to examine other potential contributing factors and identify limiting gene candidates, we plan to carry out new correlation analyses between our proteomic/transcriptomic datasets and published genome-wide datasets that report various variables under unperturbed conditions (e.g., mRNA/protein concentration, mRNA degradation rates, fitness cost, transcription/translation initiation rates, and essentiality).

Overall, this study provides a fundamental contribution to bacterial physiology by illuminating the relationship between DNA, mRNA, and protein in determining growth rate. While coarse-grained, the work invites exciting questions about how the composition of major cellular components is fine-tuned to a cell's needs and which specific gene products mediate this connection. This work has implications not only for biotechnology, as the authors discuss, but potentially also for our understanding of how DNA-targeted antibiotics limit bacterial growth.

Good point about the DNA-targeted antibiotics. Thank you!

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